Back-End APIs

HTTP, RESTful APIs, Postman, APIs with Node.js and Next.js, JWT Tokens

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SoftUni AI

https://ai.softuni.bg

Agenda

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1. **Intro to Back-End Technologies**: client-server, APIs, databases, deployment
2. **HTTP Protocol**: requests, responses, methods, status codes; **URL format**
3. **RESTful APIs and Postman**: REST principles, testing with Postman, GitHub API
4. **Minimalistic TODO List API with Node.js**: build, run, test with Postman
5. **RESTful API Design Guidelines**: resource naming, methods, status codes, JSON
6. **Next.js Basics and APIs**: pages, API route handlers, implementing REST APIs
7. **TODO List API with Next.js**: CRUD endpoints (GET, POST, PATCH, DELETE)
8. **TODO List Client App**: view / add / edit / delete TODOs in Next.js
9. **Authentication & Authorization**: JWT tokens, Bearer auth, login flow
10. **TODO List API + Client with Auth**: register, login, protected endpoints
11. **Deploy Next.js App to Netlify**: publishing process, serverless limitations

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Breaks

20:00 / 21:00

Intro to Back-End Technologies

Back-End, Client-Server, APIs, Databases, Deployment, Serverless

Client-Server Model

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🖥️ Front-End (Client)

What users see and interact with

- HTML, CSS, JS, React, Vue, Angular, Bootstrap, …
- Runs in the **Web browser**

⚙️ Back-End (Server)

Business logic, data, storage, security

- Node.js, Python, Java, C#, Go, PHP, …
- Runs on a **server** (or serverless function)

🔄 Front-End ⇄ Back-End Communication

- Over **HTTP** (or **HTTPS**)
- Typically, through **APIs** (JSON)
- Flow: **Browser** ↔ **HTTP** ↔ **Server** ↔ **Database**

What is an API?

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📡 API == Application Programming Interface

- A **contract**: how two programs talk to each other (set of rules)
- On the Web: client sends a **request** → server returns a **response**

🧩 Common API Styles

- **REST** (RESTful APIs) – resource-based, uses HTTP verbs
- **GraphQL** – single endpoint, query language for data
- **gRPC** / **WebSockets** / **SOAP** – less common in Web apps

👉 In this lesson: we focus on **RESTful APIs**

Databases and ORM Frameworks

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🗄️ Databases

Store app data persistently

- **Relational**: PostgreSQL, MySQL, SQLite, SQL Server
- **Non-relational**: MongoDB, Redis, DynamoDB

🔗 ORM Frameworks

Object-Relational Mapping

- Map DB table rows to **objects** in the code
- Node.js ORMs: **Prisma**, **Drizzle**, **TypeORM**, **Sequelize**

👉 In this lesson: we keep data in a **simple JSON file**

No DB – focus on **HTTP**, **REST**, and **APIs** first

Deployment, Cloud, Containers

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🌐 Where Does the Back-End Run?

- **Traditional servers**: VPS, dedicated hosts (DigitalOcean, Hetzner)
- **Cloud platforms**: AWS, Azure, Google Cloud
- **PaaS**: Netlify, Vercel, Railway, Render
- **Containers**: Docker + Kubernetes

🛠️ Two Common Hosting Styles

- **Server-based**: a Node.js process runs 24/7 (takes resources)
- **Serverless**: code runs on-demand, auto-scales, no server to manage

Intro to the HTTP Protocol

Requests, Responses, Methods, Status Codes

HTTP Basics

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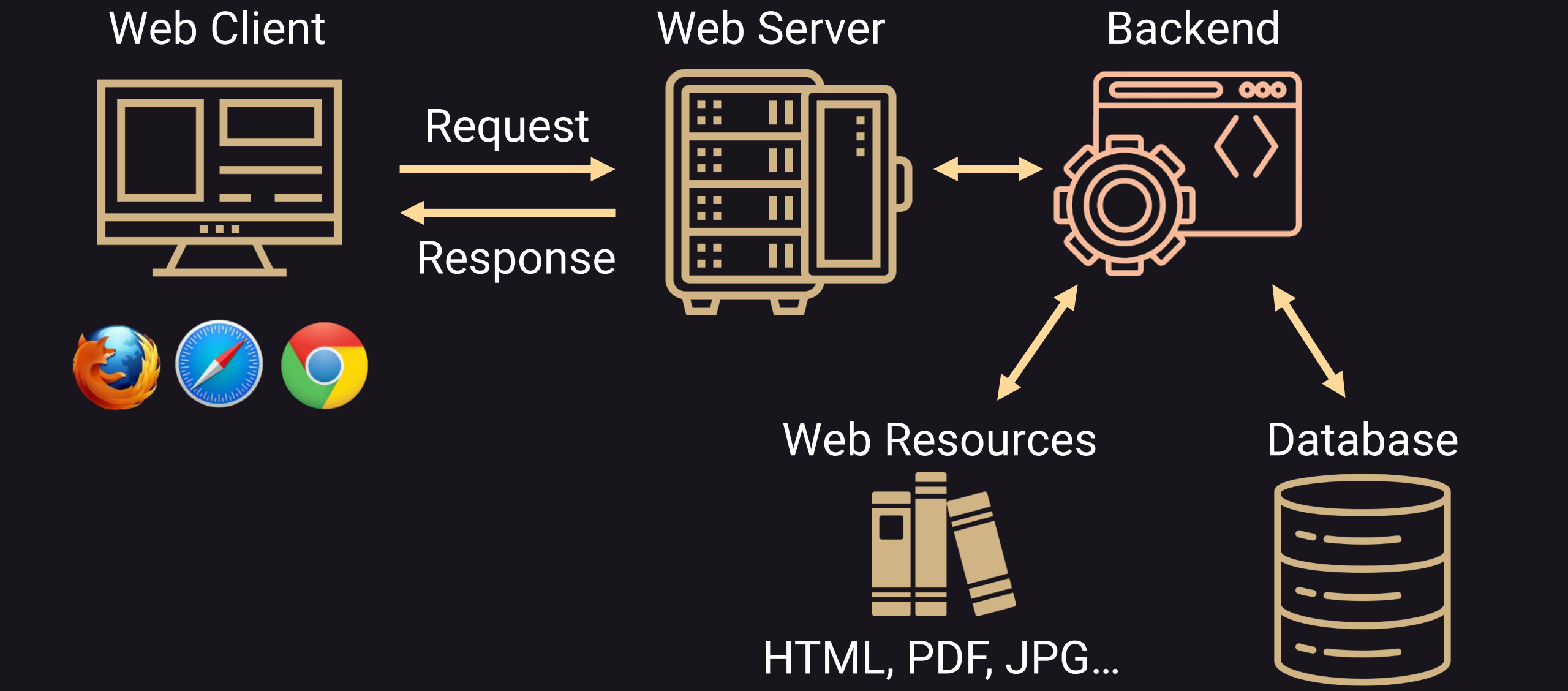
- **HTTP** (**H**yper**T**ext **T**ransfer **P**rotocol)
  - Text-based **client-server protocol** for the Internet
  - For transferring Web **resources**: HTML, images, styles, etc.
  - **Request-response** based, relies on URLs (like https://softuni.org)
  - **Stateless** (no memory between requests)

The Client-Server Model in Web Apps



HTTP Request Format

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📥 HTTP Request Parts

- **Request line**
  - HTTP method + path + version
  - *e.g.* **GET /api/todos HTTP/1.1**
- **Headers**
  - Metadata as key-value pairs
  - **Host**, **Content-Type**, **Authorization**, **User Agent**
- **Body** (optional)
  - JSON, form data, file, …

Example – HTTP POST Request

```
POST /api/todos HTTP/1.1
Host: example.com
Content-Type: application/json
Authorization: Bearer eyJhbGciOi…

{ "title": "Learn JS and React",
"completed": false }
```

HTTP Request Methods

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- **HTTP** defines **methods** to indicate the desired action to be performed on the identified resource

Method	Description
GET 	Retrieve a resource
POST 	Create / store a resource
PUT 	Update (replace) a resource
DELETE 	Delete (remove) a resource
PATCH 	Update resource partially (modify)
HEAD 	Retrieve the resource's headers

CRUD == the four main functions of persistent storage

Method
CONNECT
OPTIONS
TRACE

HTTP Response Format

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HTTP Response Parts

- **Status line**
  - Version + status code + status text
  - *e.g.* **HTTP/1.1 201 Created**
- **Headers**
  - Metadata about the response
  - **Content-Type**, **Content-Length**, **Set-Cookie**
- **Body**
  - The returned **data** — HTML / JSON / image / script / …

Example – HTTP Response

```
HTTP/1.1 201 Created
Content-Type:
application/json
Content-Length: 62
Set-Cookie: lang=en; Path=/

{ "id": 7, "title": "Learn JS
and React", "completed":
false }
```

HTTP Response Status Codes

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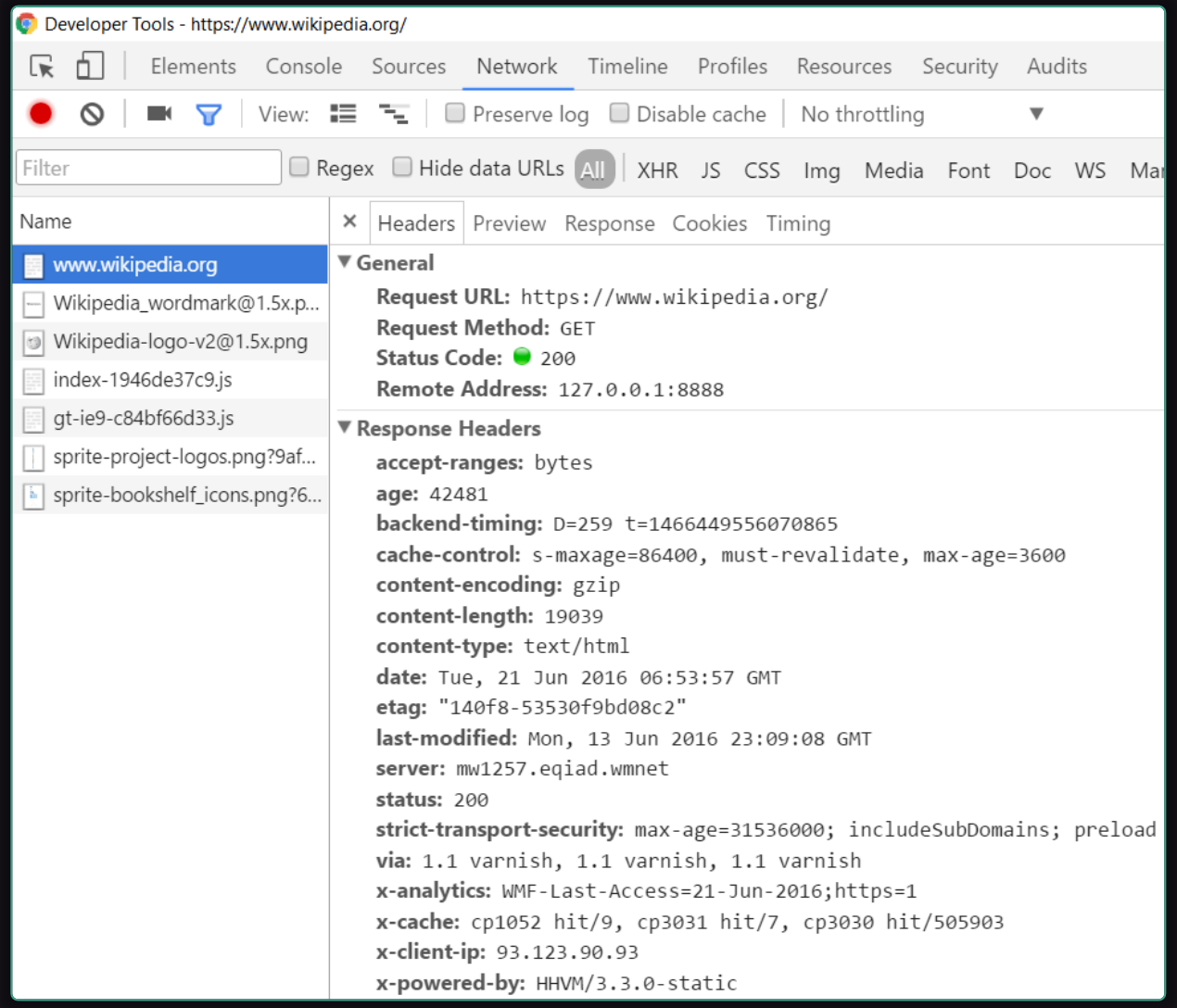
Status Code	Action	Description
200	OK	Successfully retrieved resource
201	Created	A new resource was created
204	No Content	Request has nothing to return
301 / 302	Moved	Moved to another location (redirect)
400	Bad Request	Invalid request / syntax error
401 / 403	Unauthorized	Authentication failed / Access denied
404	Not Found	Invalid resource was requested
409	Conflict	Conflict was detected, e.g. duplicated email
500 / 503	Server Error	Internal server error / Service unavailable

Browser Dev Tools: Network Inspector



- **Chrome Dev Tools**
  - Press **[F12]** in Chrome
  - Open the [Network] tab
  - Inspect the HTTP traffic

Example: Fetch Existing URL (200 OK)

- Use the Web browser, **Postman**, or **curl** to send this **request**:

```
GET https://api.github.com/users/nakov
```

- The HTTP server **responds** with the requested resource:

HTTP response (200 OK)

```
HTTP/1.1 200 OK
Content-Type: application/json

{ "login": "nakov", "id": 1165408, "public_repos": 57, … }
```

Example: Fetch Non-Existing URL (404)

- Request a resource that **does not exist**:

```
https://api.github.com/users/no-such-user72739
```

- The HTTP server replies with **404 Not Found**:

HTTP response (404 Not Found)

```
HTTP/1.1 404 Not Found
Content-Type: application/json

{ "message": "Not Found", "documentation_url": "…" }
```

URLs and URL Format

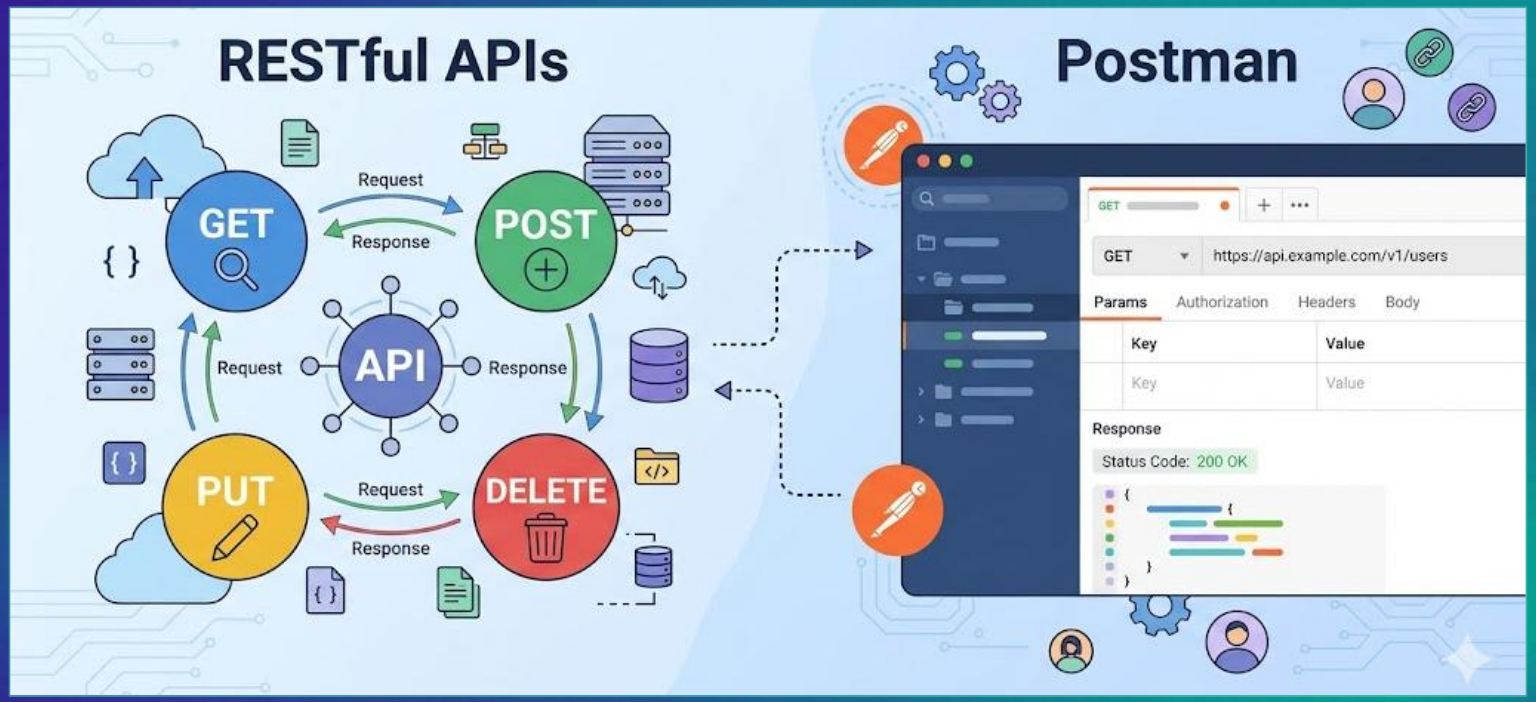
Protocol, Host, Path, Query String

Uniform Resource Locator (URL)

- Network **protocol** (**http**, **ftp**, **https**, …) – HTTP in most cases
- **Host** or **IP address**, e.g. **gmail.com**, **127.0.0.1**, **websrv**
- **Port** (the default port is **80**) – integer in the range [0…65535]
- **Path**, e.g. **/news**, **/products/7**, **/scripts/index.php**
- **Query string**, e.g. **?id=27&lang=en**
- **Fragment**, e.g. **#slides** – a HTML section in the page

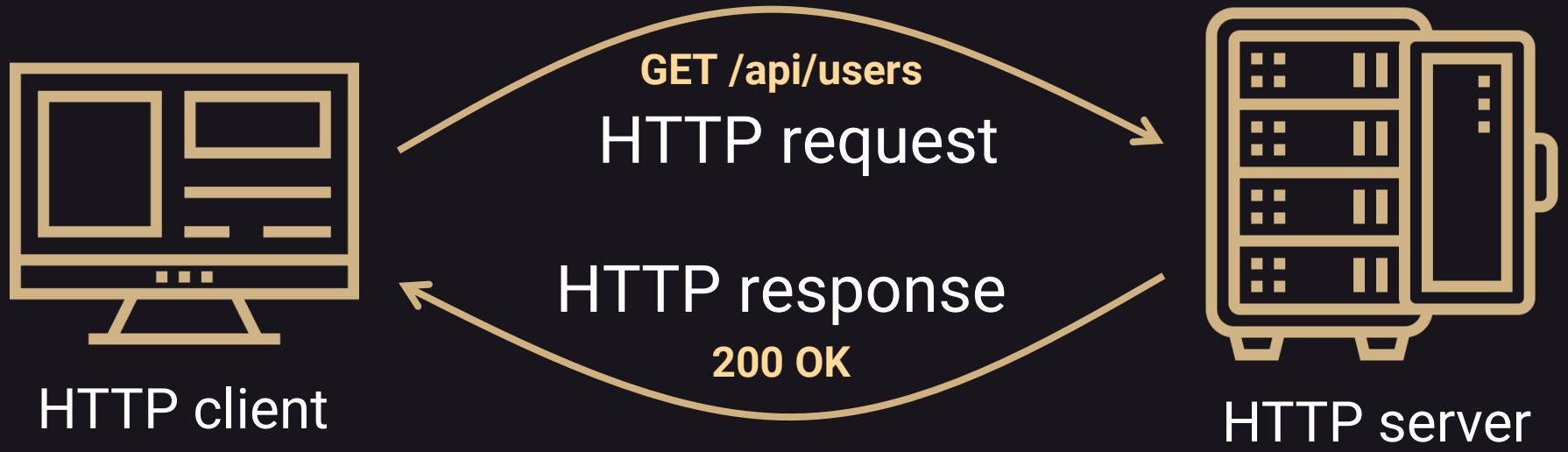
RESTful APIs and Postman

REST Principles, Playing with RESTful APIs with Postman, Example: GitHub API



RESTful APIs

- **RESTful APIs** are HTTP-based backend APIs (Web services)
  - HTTP methods **GET**, **POST**, **PATCH** and **DELETE** implement the **CRUD** operations (retrieve, create, modify and delete data)



RESTful APIs: Principles

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- A **RESTful API** exposes server resources via **URLs** and **HTTP**
  - Example of server resource: **/api/users/42**
- **HTTP method** == the **action** (GET → read, POST → create, PUT/PATCH → update, DELETE → remove)
- Data is typically **JSON** (set Content-Type: application/json)
- **Status codes** tell what happened (200 → OK, 201 → Created, 401 → Unauthorized, 404 → Not Found, 500 → Server Error)
- **Stateless** – no sessions; send auth token with every request
- **Endpoints** are **nouns** (not verbs): **/api/users** (not **/getUsers**)

Example – GitHub API – List Repos

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List Public GitHub Repos

For any GitHub user — **no auth needed**

- **Response**: JSON array of repo objects

Try in the browser:

- Just open the URL
- HTTP GET can be sent directly

Example: List Repos

```
GET https://api.github.com
/users/nakov/repos
```

```
// Response (truncated)
[
  { "name": "softuni-ai",
    "stargazers_count": 42 },
  …
]
```

Postman: HTTP Testing Tool for Devs

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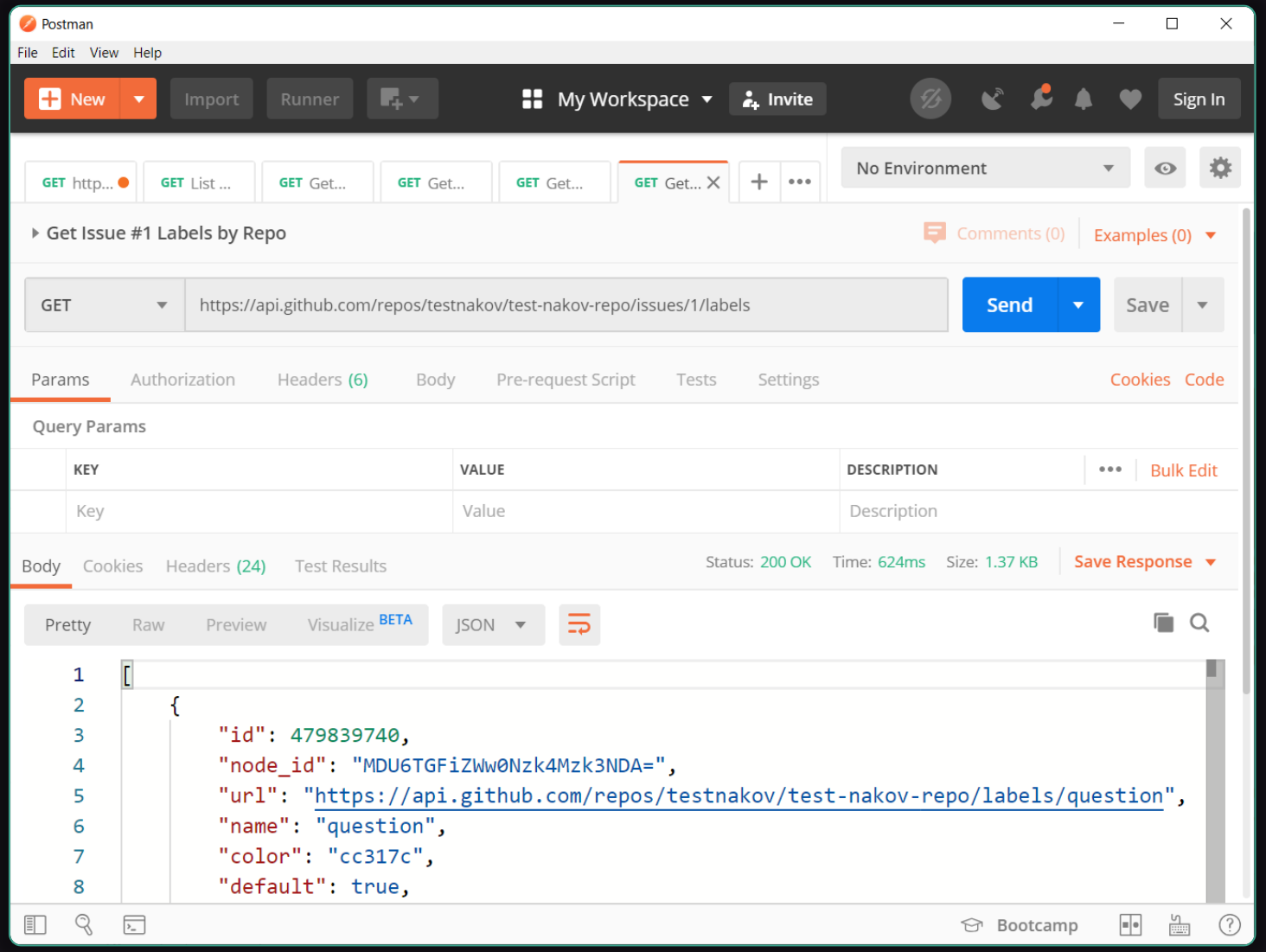
Postman

- HTTP client tool for developers
- Compose and send **HTTP requests**
- Alternatives
  - Insomnia, SoapUI, Hoppscotch



Example – GitHub API with Postman

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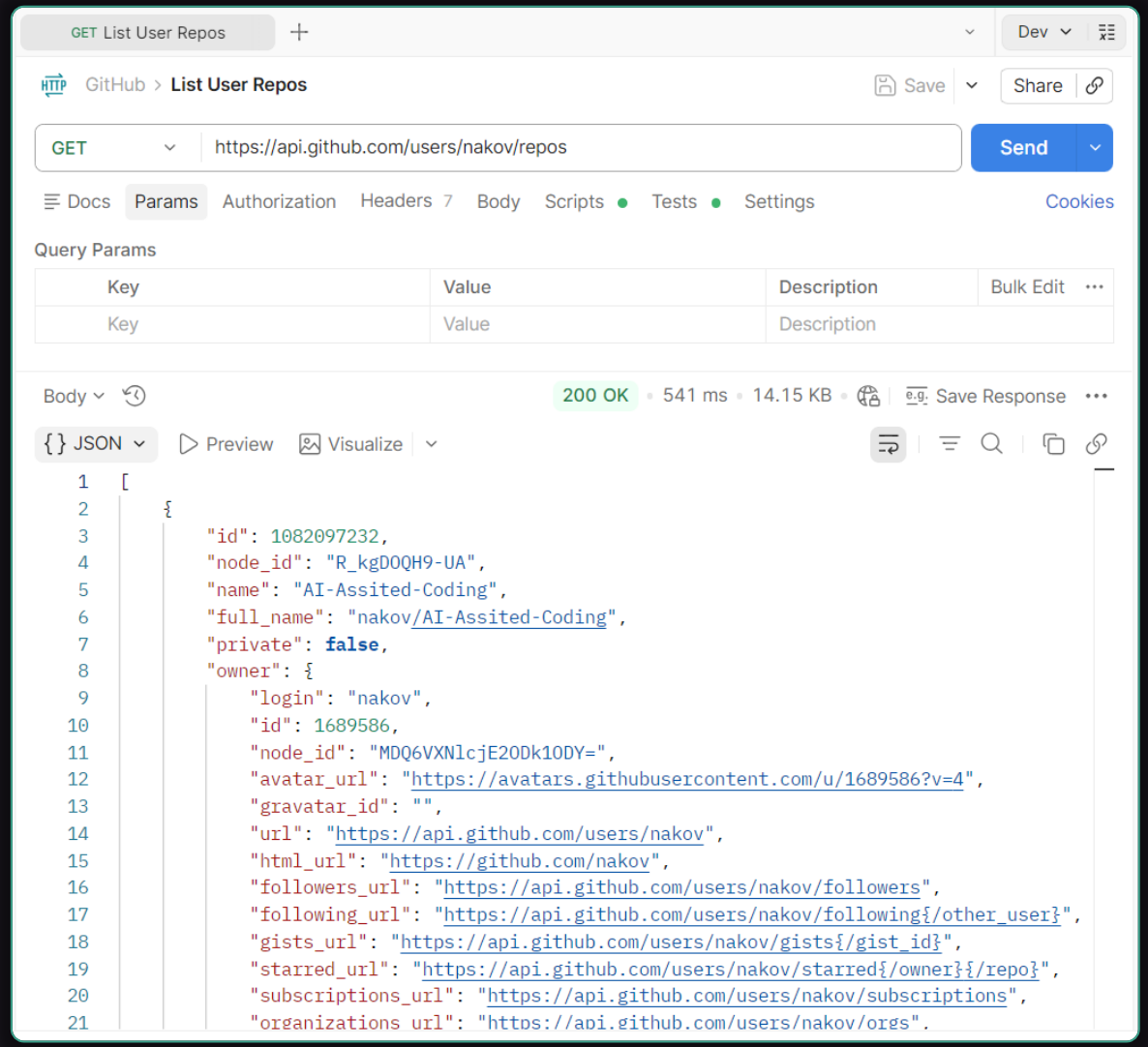
- Try the "List GitHub Repos" request in **Postman**:
  - Method: **GET**
  - URL: **https://api.github.com/users/{user}/repos**
  - Click [Send] → inspect the HTTP response



Example — Minimalistic HTML + JS App

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- **AI prompt** to generate a GitHub repos viewer:

```
Create a minimalistic GitHub Repo Viewer in HTML + JS:
  - Input field for a GitHub username
  - Button [Load Repos]
  - On click -> fetch
https://api.github.com/users/{username}/repos
  - Display the list of repos with name + link
  - Use plain JS (fetch API), no frameworks, single HTML file
```

📸 Try It Live

Paste prompt into AI → run in browser → test with any GitHub username

🔗 Endpoint Used

GET
https://api.github.com/users/{username}/repos

GitHub Repo Viewer App

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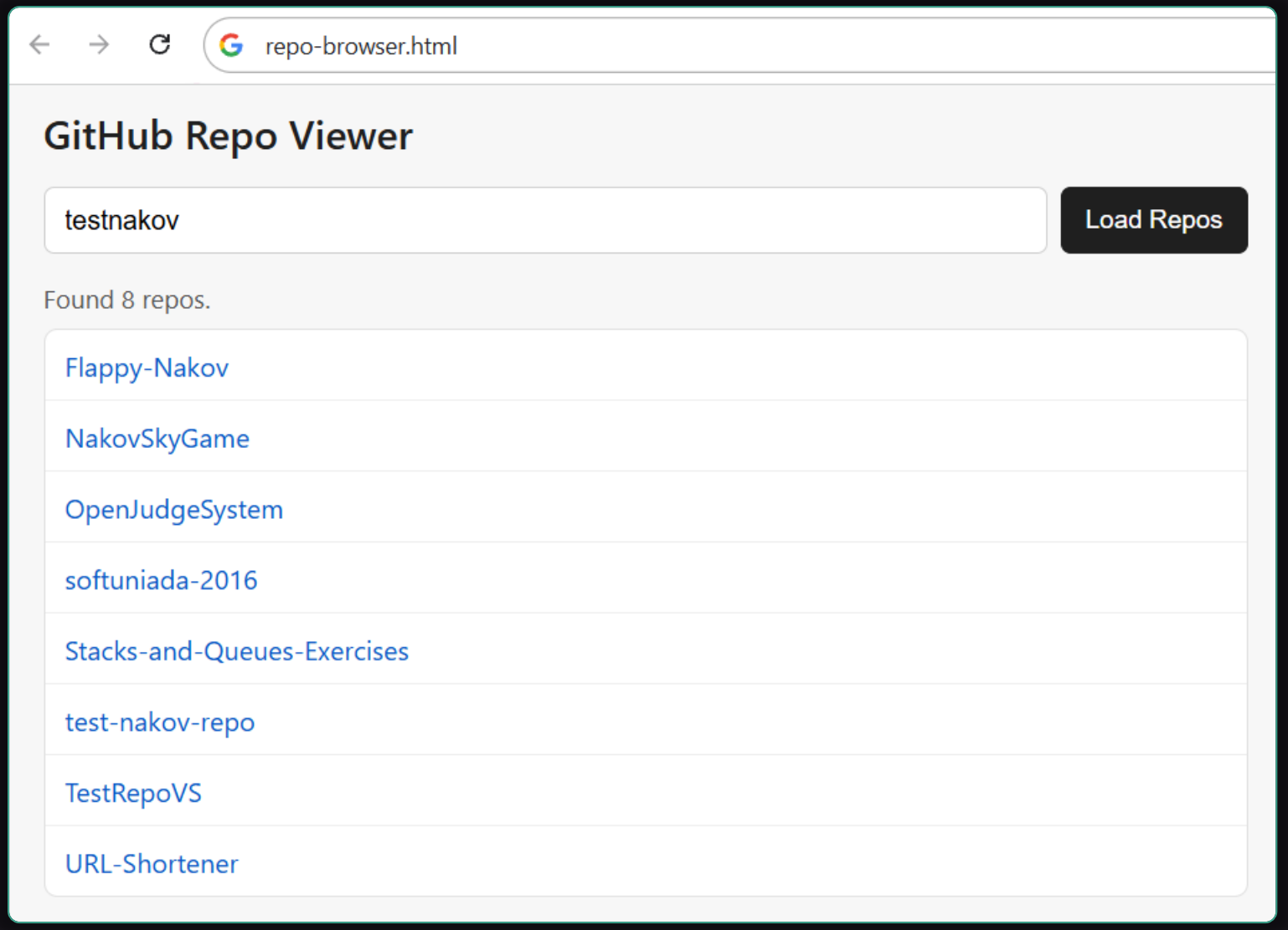

Creating an API Key in GitHub

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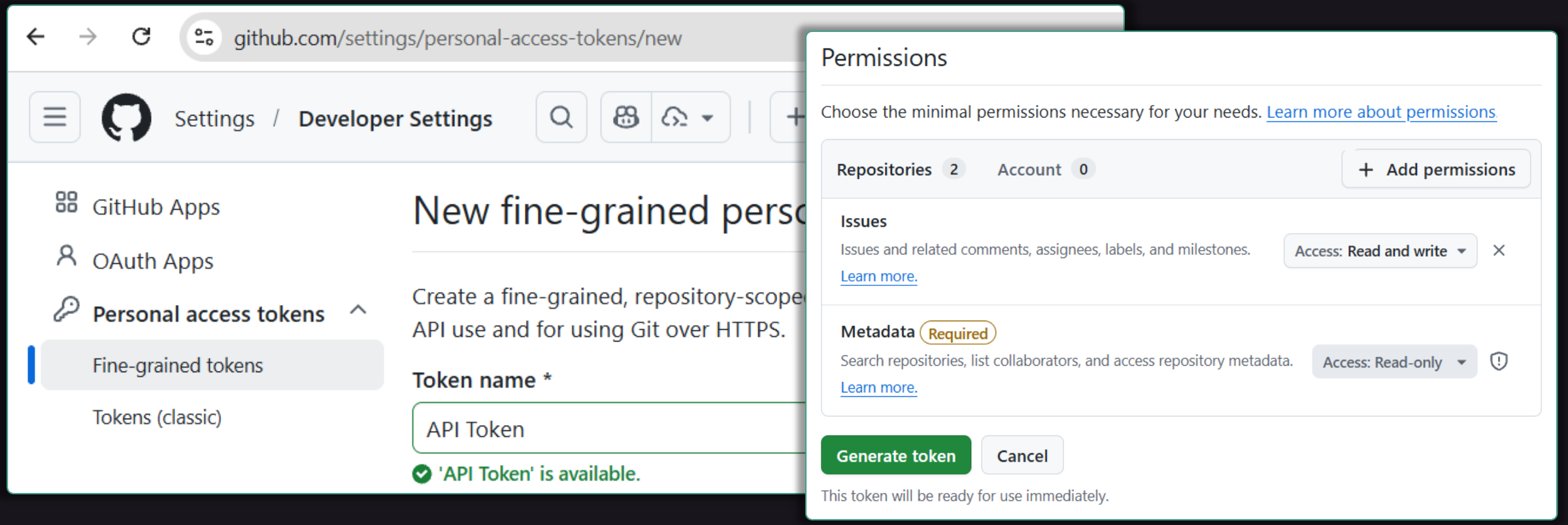
- **API keys** in GitHub:
  - https://github.com/settings/personal-access-tokens



RESTful APIs with Authentication

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🔑 Why Authentication and Authorization?

- Many APIs require an **auth token** to verify who you are
- **Personal Access Token (PAT)** in the `**Authorization**` header

🔑 Authorization Flow

- **Login** → returns JWT token
- **Next request** → send the token in the `**Authorization**` header

📡 Create a GitHub Issue (POST)

```
POST
https://api.github.com/repos/testnakov/test-nakov-repo/issues
Authorization: Bearer github_…
Content-Type: application/json
Body: { "title": "Bug: login fails", "body": "Steps …" }
```

```
201 Created + new issue JSON
```

🔑 **Keep tokens secret**. Never commit to Git!

Example — List / Create Issues App

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- **AI prompt** to generate a GitHub Issues Browser app:

```
Create a minimalistic GitHub Issues browser in HTML + JS:
  - Input: GitHub repo (owner/name), auth token (PAT)
  - Button [List Issues] -> GET /repos/{owner}/{repo}/issues
  - Form: title + body + [Create Issue] button
  - POST /repos/{owner}/{repo}/issues with Bearer token
  - Keep it simple: single HTML file (HTML + CSS + JS)
```

✅ What This App Demonstrates?

- Authenticated GET request — listing existing issues
- Authenticated POST request — **creating a new issue** with JSON body
- Real-world GitHub API workflow with **token auth**

GitHub Issue Browser App

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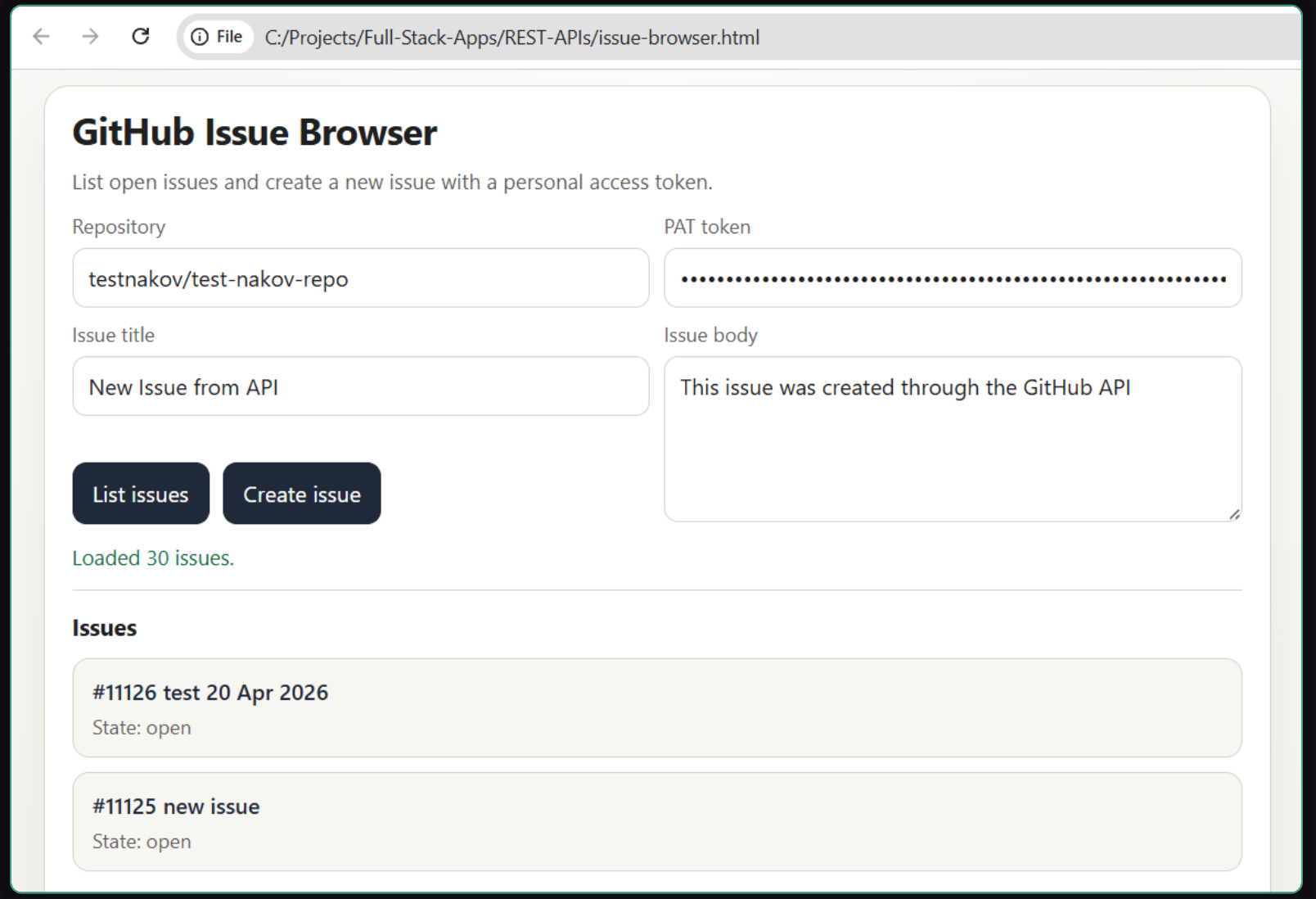
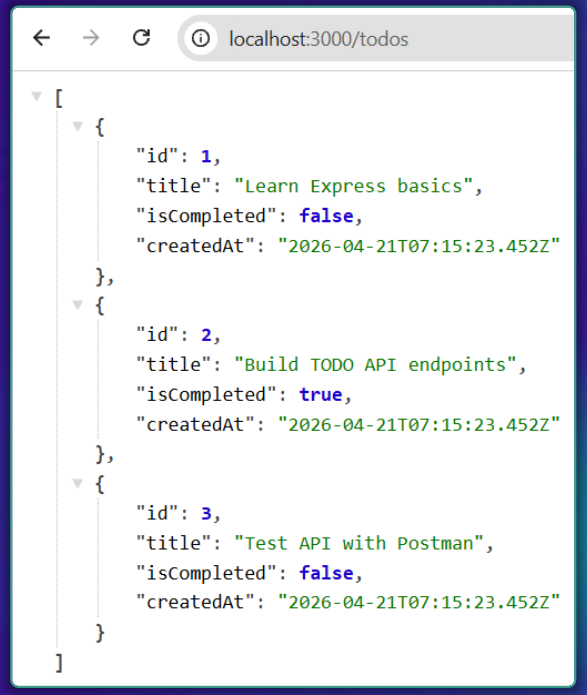
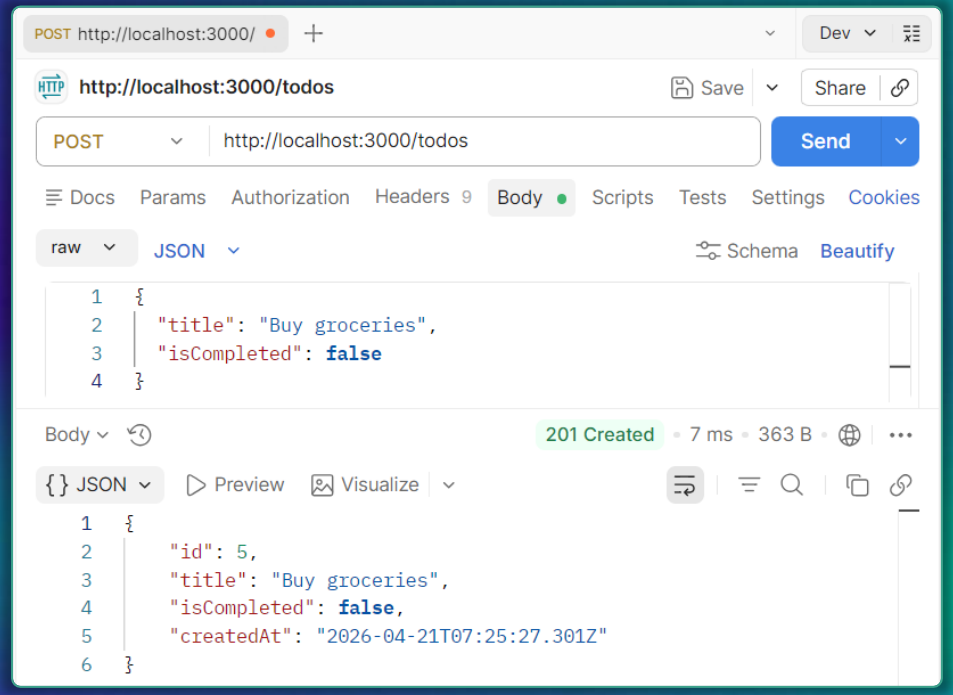
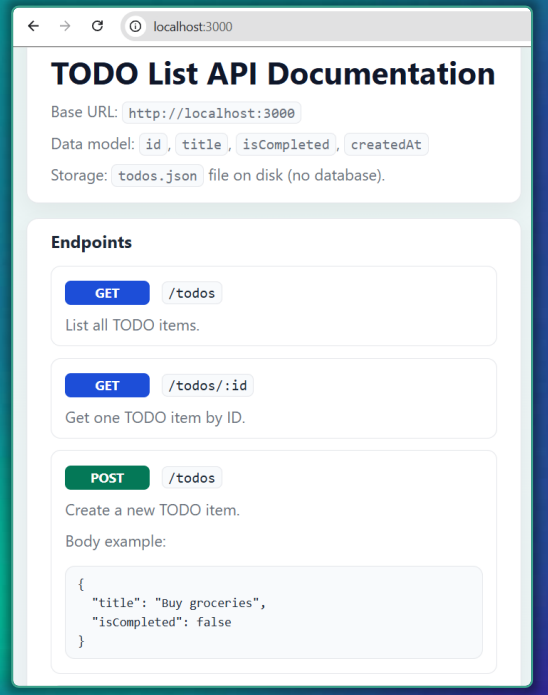


TODO List API in Node.js

Building a RESTful API with Node.js + Express, Testing with Postman







Creating a RESTful API Server

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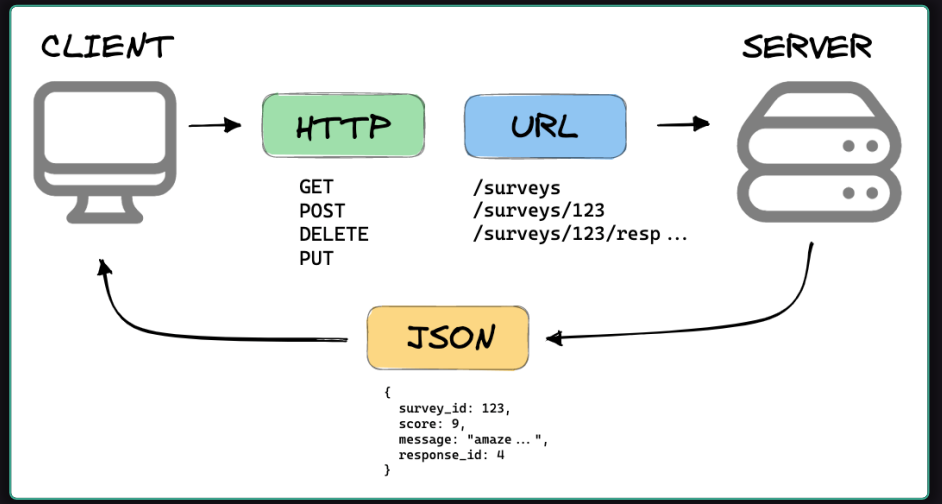
- Creating a RESTful API server in JavaScript
  1. Implement **API route handlers**:
     - Handle **GET /endpoint**
     - Handle **POST /endpoint**
  2. Listen to certain **port**, e.g. 8080



- Implementations:
  - **Node.js** + **Express**: simple, straightforward, flexible
  - **Next.js** API route handlers: established app framework

RESTful API Server in Node.js – Example

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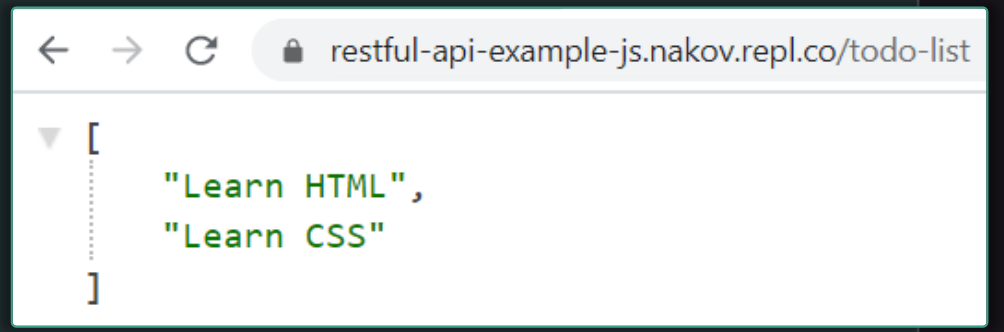
```
const express = require('express');
const app = express();
app.use(require('cors')());
app.use(require('body-parser').json());

let todoList = ['Learn HTML', 'Learn CSS'];

app.get('/todo-list', (req, res) => {
  res.json(todoList);
});

app.post('/todo-list', (req, res) => {
  todoList.push(req.body.todoText);
  res.sendStatus(200);
});

app.listen(8080);
```

```
npm install express cors
```

```
node ./api-server.js
```



🌐 **Minimalistic Demo**

- Note: this is a minimalistic demo: not for real use

TODO List API — Scope

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- Implement a **TODO List API** in Node.js + Express

🗄️ Data Model

- **id** – unique number
- **title** – string
- **completed** – boolean
- **createdAt** – ISO date

🌐 API Docs @ Home Page

- Displays an HTML docs
- Lists all endpoints

📋 API Endpoints

- **GET /todos** – list all TODOs
- **GET /todos/:id** – get one TODO
- **POST /todos** – create
- **PATCH /todos/:id** – update
- **DELETE /todos/:id** – delete

💾 Storage Model

- Plain JSON file: `**todos.json**`
- No database – focus on HTTP & REST first

TODO List API — AI Prompt

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- **AI prompt** to generate a TODO List API:

```
Implement a simple TODO List API in Node.js + Express:
  - Data model: id, title, isCompleted, createdAt
  - Storage model: JSON file: `todos.json` (no database)
  - Initially, load sample data (on first run)
  - API endpoints:
    - GET /todos – list all TODOs
    - GET /todos/:id – get one TODO
    - POST /todos – create
    - PATCH /todos/:id – update
    - DELETE /todos/:id – delete
  - At the home page implement HTML docs: list all endpoints
```

TODO List API — The App

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localhost:3000/todos


```
[
  {
    "id": 1,
    "title": "Learn Express basics",
    "isCompleted": false,
    "createdAt": "2026-04-21T07:15:23.452Z"
  },
  {
    "id": 2,
    "title": "Build TODO API endpoints",
    "isCompleted": true,
    "createdAt": "2026-04-21T07:15:23.452Z"
  },
  {
    "id": 3,
    "title": "Test API with Postman",
    "isCompleted": false,
    "createdAt": "2026-04-21T07:15:23.452Z"
  }
]
```


localhost:3000

TODO List API Documentation

Base URL: `http://localhost:3000`

Data model: `id`, `title`, `isCompleted`, `createdAt`

Storage: `todos.json` file on disk (no database).

Endpoints

GET `/todos`

List all TODO items.

GET `/todos/:id`

Get one TODO item by ID.

POST `/todos`

Create a new TODO item.

Body example:

```
{
  "title": "Buy groceries",
  "isCompleted": false
}
```

POST http://localhost:3000/

http://localhost:3000/todos

Save | Share

POST | http://localhost:3000/todos | Send

Docs | Params | Authorization | Headers 9 | Body | Scripts | Tests | Settings | Cookies

raw | JSON | Schema | Beautify

```
{
  "title": "Buy groceries",
  "isCompleted": false
}
```

Body | 201 Created | 7 ms | 363 B

{} JSON | Preview | Visualize

```
{
  "id": 5,
  "title": "Buy groceries",
  "isCompleted": false,
  "createdAt": "2026-04-21T07:25:27.301Z"
}
```

Running the API Server

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🗂️ Project Files Generated

- **server.js** — main entry point
- **todos.json** — JSON storage file (app data)
- **package.json** — npm scripts + dependencies
- **docs.html** — generated endpoint docs

▶️ Start the Server

- **npm install**
- **npm start**
- Output: *Server running at http://localhost:3000*

🔍 Quick Smoke-Tests in the Browser

- Open browser: http://localhost:3000/ → HTML docs page listing all endpoints
- GET http://localhost:3000/todos → JSON array of sample TODOs (200 OK)
- GET http://localhost:3000/todos/1 → single TODO object (200 OK)
- GET http://localhost:3000/todos/999 → { "error": "Not found" } (404 Not Found)

Testing with Postman

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🧪 Test the TODO API with Postman

- Create **Postman collection**: "TODO API"
- **GET** http://localhost:3000/todos → 200 OK, JSON array
- **GET** http://localhost:3000/todos/1 → 200 OK + TODO object
- **GET** http://localhost:3000/todos/999 → 404 Not Found + error JSON
- **POST** http://localhost:3000/todos + JSON body → 201 Created + TODO object
- **POST** http://localhost:3000/todos + empty body → 400 Bad Request
- **PATCH** http://localhost:3000/todos/1 + partial JSON → 200 OK, updated TODO
- **PATCH** http://localhost:3000/todos/999 + partial JSON → 404 Not Found
- **DELETE** http://localhost:3000/todos/1 → 200 OK or 204 No Content
- **DELETE** http://localhost:3000/todos/999 + partial JSON → 404 Not Found

RESTful API Design Guidelines

Resource Naming, HTTP Methods, Status Codes, JSON, Filtering, Pagination

Resource Naming

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📌 Use Nouns, Not Verbs

- ✅ **GET /users, POST /users**
- ❌ **GET /getUsers**
- Use plural nouns:
  /products, **/orders**, **/todos**
- Use lowercase + hyphens:
  /blog-posts (not **/blogPosts**)

🔑 Key Naming Rules

- Collections → plural nouns:
  /todos, **/users**, **/repos**
- Single resource → noun + ID:
  /todos/7, **/users/42**
- Sub-resources:
  /posts/5/author, **/posts/5/visibility**

🔗 Nested URLs for Relations

- **/users/5/orders** → orders of user #5
- **/users/5/todos** → TODOs of user #5
- Keep nesting shallow — max 2 levels deep

HTTP Methods Mapping

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🔁 CRUD → HTTP Mapping

- Create → **POST /users**
- Read (list) → **GET /users**
- Read (one) → **GET /users/5**
- Read (report) → **GET /users/stats**
- Update (partial) → **PATCH /users/5**
- Update (full) → **PUT /users/5**
- Update (state) → **PUT /posts/5/visibility → { "hidden": true }**
- Delete → **DELETE /users/5**

✅ GET Is Always Safe

- Never modifies data — safe to retry
- Idempotent: same result every call

⚠️ Modifying methods

- **POST** — creates; not idempotent
- **PATCH** — partial update
- **PUT** — replaces the entire object
- **DELETE** — removes; idempotent

HTTP Status Codes Guidelines

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✅ 2xx Success

- **200 OK** — success
- **201 Created** — after POST
- **204 No Content** — after DELETE
- **202 Accepted** — async job started
- **206 Partial Content** — bytes range

⚠️ 4xx Client Errors

- **400 Bad Request** — invalid input
- **401 Unauthorized** — no auth / bad auth
- **403 Forbidden** — action not allowed
- **404 Not Found** — missing resource
- **409 Conflict** — duplicate (on create / edit)

🔥 5xx Server Errors

- **500 Internal Server Error** — unhandled failure / exception
- **503 Service Unavailable** — server overloaded
- **502 Bad Gateway** / **504 Gateway Timeout** – proxy failed to access upstream
- **501 Not Implemented** — functionality unavailable

JSON Responses

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📦 Always Return JSON for Data Endpoints

- Set **Content-Type: application/json** on all responses
- Keep **response shape** consistent — same fields, same types
- Use **camelCase** for field names: **createdAt**, **userId**, **isActive**

📄 Single Resource

```
{ "id": 1, "title": "Sleep", "completed": false }
```

📋 Collection

```
[ { "id": 1, ... }, { "id": 2, ... } ]
```

❌ Error Response

```
{ "error": "TODO not found", "code": 404 }
```

✅ Best Practices

- Wrap lists in object: { data: [...], total: 42 }
- Include error details in error responses

Pagination, Filtering and Versioning

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📄 Pagination

- **GET /api/v1/users?page=2&pageSize=20**
- Prevents returning 10 000 items at once
- Return **data** + **count**: { data: [...], total: 847 }

🔀 Versioning

- Keep old clients working when API changes
- **/api/v1/users** → **/api/v2/users**

🔍 Filtering & Sorting

- Filter: **GET /users?role=admin&active=true**
- Sort: **GET /users?sort=createdAt&order=desc**
- Combined: **?role=admin&sort=name&page=1**

💡 Full Example

- GET /api/v1/users
  - ?role=admin
  - &active=true
  - &sort=createdAt&order=desc
  - &page=2&pageSize=20

Next.js Basics and APIs

Next.js Apps Structure, API Route Handlers

What is Next.js?

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⚡ Next.js — React-Based Full-Stack Framework

- Built on top of **React** by **Vercel**
- Used by Netflix, TikTok, Nike, and many more
- https://nextjs.org

📁 Key Features

- File-based **routing** (pages in **app/** folder)
- **Server** and **client** components
- Built-in **API routes** — back-end in same project
- **Server-side rendering** (SSR)
- Image optimization, built-in fonts, middleware

🚀 Why Next.js for Back-End APIs?

- One project, one deploy — **front-end + back-end** together
- API routes live at **app/api/*/route.ts** — standard file-based routing
- Works seamlessly on **Vercel**, **Netlify**, **Railway** — serverless or server-based
- **TypeScript** first-class – great developer experience

Create a Next.js Project

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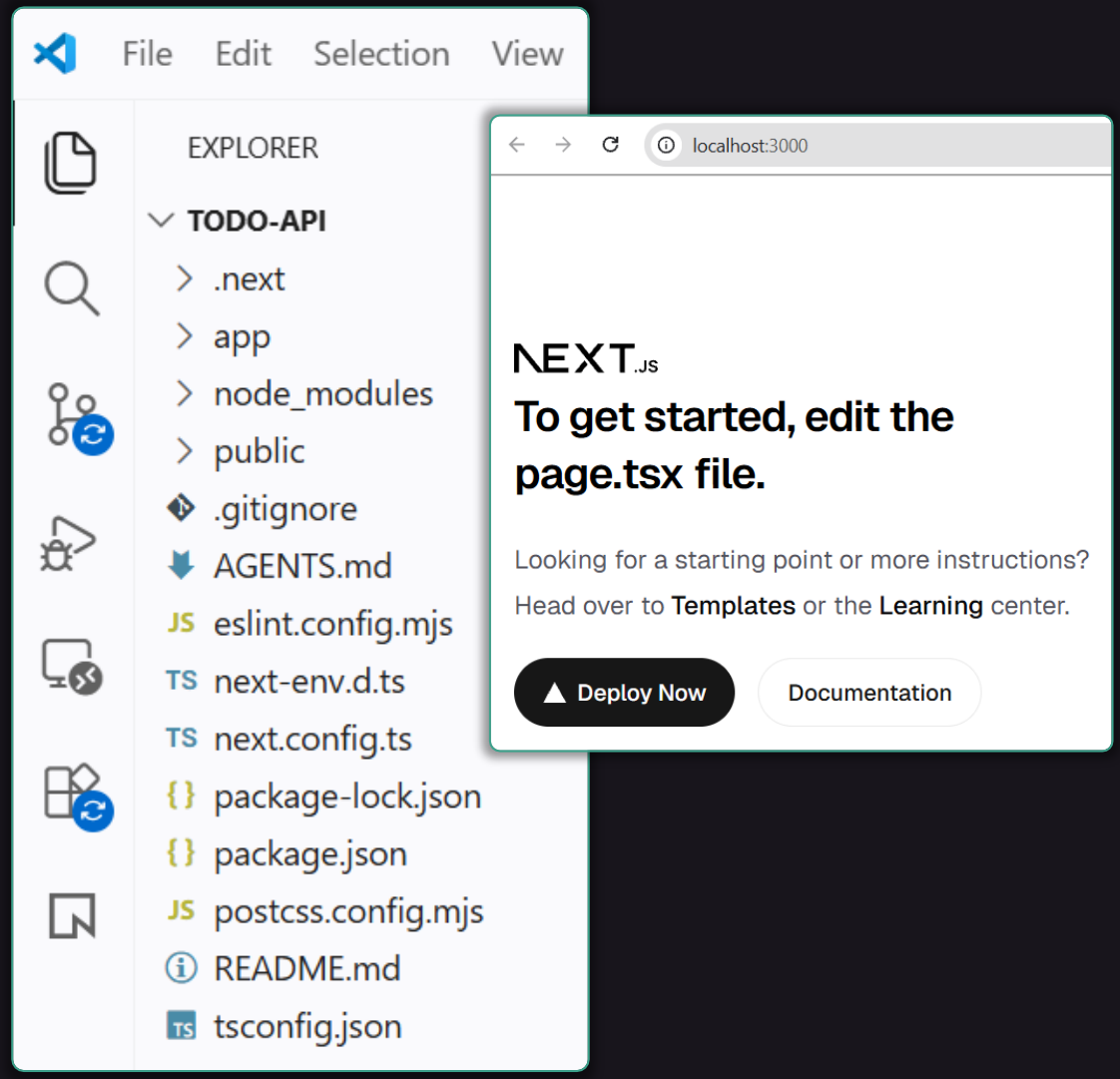
🚀 Initialize a Next.js Project

```
npx create-next-app@latest todo-api
```

- Answer the prompt:
  - *Yes, use recommended defaults*

▶️ Run the `dev` Server

- **npm run dev**
- Open: http://localhost:3000
- Hot reload — changes appear instantly



Next.js Project Structure

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📁 Project Structure Overview

- **app/page.tsx** — home page — **/**
- **app/layout.tsx** — root layout wrapping all pages
- **app/api/** — all API route handlers live here
- **public/** — static assets (images, fonts, icons)

📁 API Route Handlers

- **app/api/hello/route.ts** — API route handlers — maps to **/api/hello**

API Route Handlers

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📡 Route Handlers in Next.js

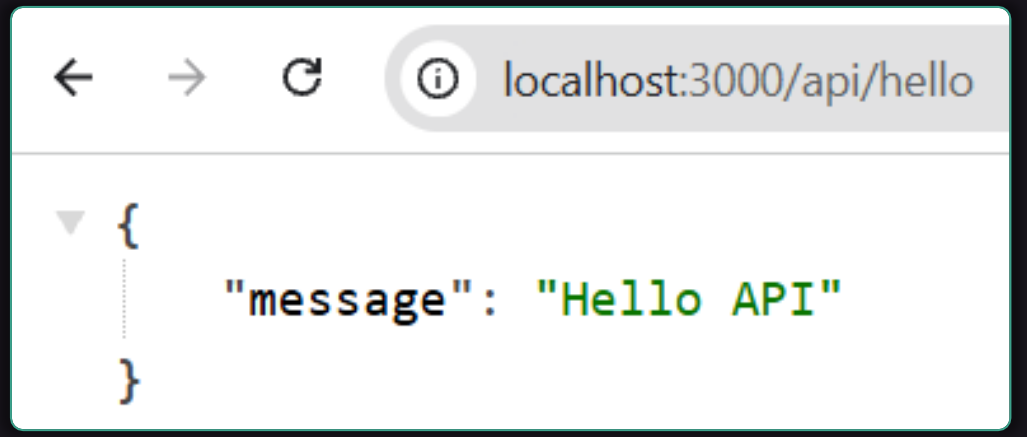
- Live in **app/api/.../route.ts**
- Export HTTP methods as named functions:
  - **export async function GET() { ... }**
  - **export async function POST(req) { ... }**
- Return **Response.json({ data })**

🔧 Accessing Request Data

- URL params: **function GET(req, { params })**
- Query string: **new URL(req.url).searchParams**
- Request body: **const body = await req.json()**
- Headers: **req.headers.get('Authorization')**

app/api/hello/route.ts

```
export async function GET() {
  let result = {"message":
  "Hello API"};
  return Response.json(result);
  // 200 OK
}
```



Example — Simple API Endpoint

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💡 AI Prompt

- Create API route: **GET /api/hello**
  - Returns **{ msg: 'Hello, <name>!' }**
  - Read **name** from query param **?name=Svetlin**
  - Default to '**API**' if not provided

📄 app/api/hello/route.ts

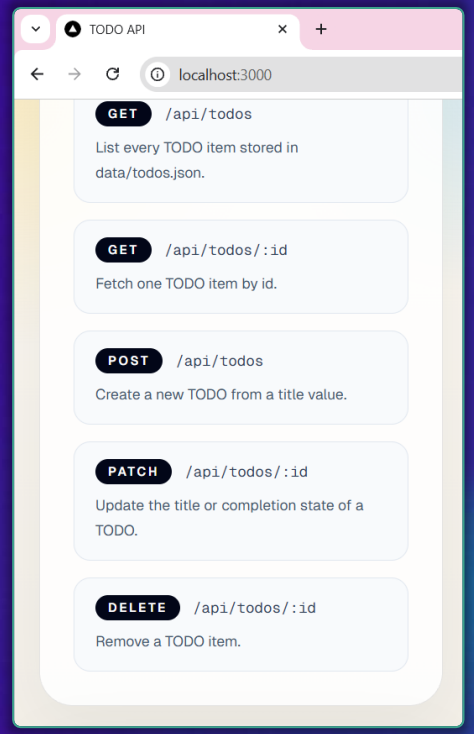
```
export async function GET(req: Request) {
  const url = new URL(req.url);
  const name = url.searchParams.
  get('name') ?? 'API';
  return Response.json({ msg: `Hello,
  ${name}!` });
}
```

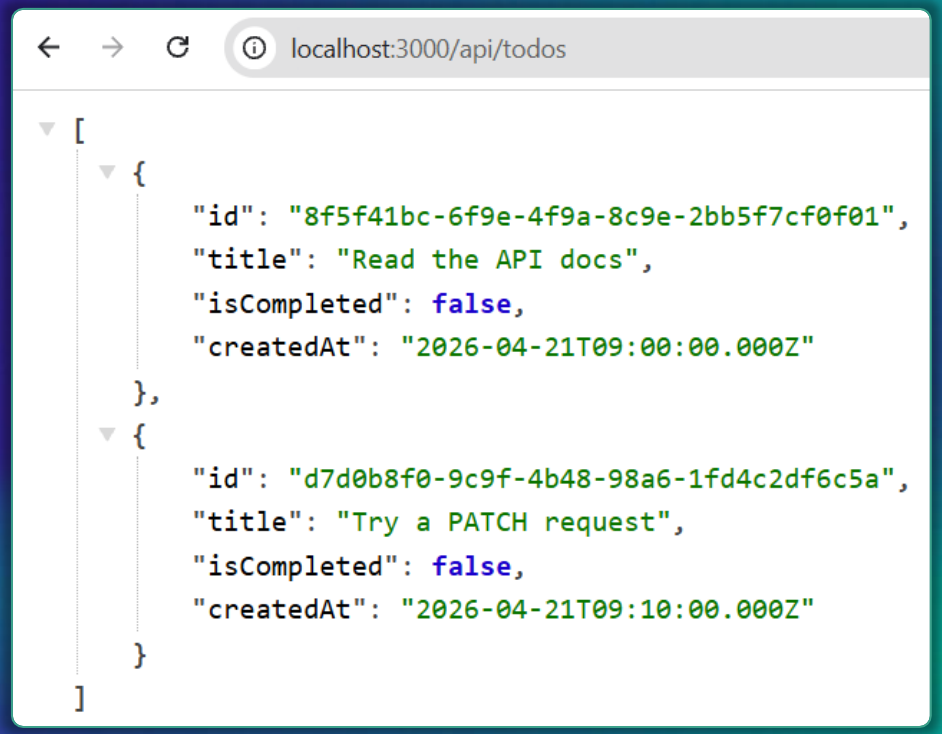
▶️ Test the API Endpoint: /api/hello

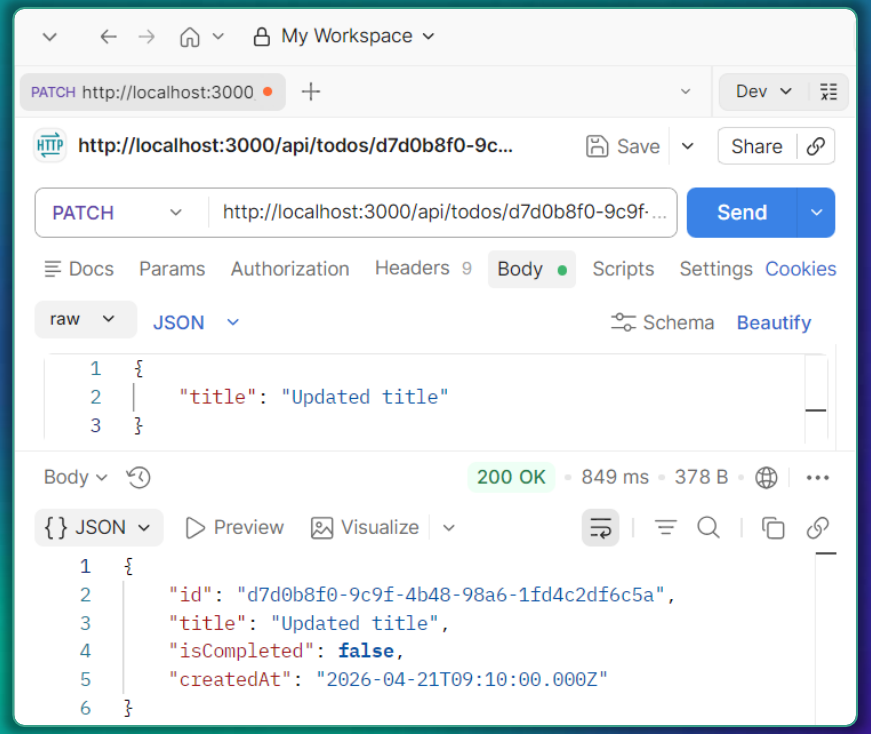
- http://localhost:3000/api/hello → { message: 'Hello, API!' }
- http://localhost:3000/api/hello?name=SoftUni → { message: 'Hello, SoftUni!' }

TODO List API with Next.js

Implementing CRUD Endpoints with Next.js







TODO API with Next.js: The Plan

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📋 Rebuild the TODO API with Next.js

- Same **CRUD endpoints** — now as **Next.js API route handlers**
- **GET / POST /api/todos** and **GET / PATCH / DELETE /api/todos/[id]**

🗂️ Project Structure

- **app/api/todos/route.ts** (GET, POST)
- **app/api/todos/[id]/route.ts** (GET, PATCH, DELETE)
- **lib/todos.ts** — data access helper
- **data/todos.json** — storage file

💾 Storage

- **data/todos.json** (no DB yet)
- **Todo { id, title, isCompleted, createdAt }**
- Initially load **sample data**

🏠 **App Home Page**: show simple API docs (describe the endpoints)

TODO List API — AI Prompt

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- **AI prompt** to generate a TODO List API in Next.js:

```
Implement a simple TODO List API with Next.js:
  - Data model: id (number), title, isCompleted, createdAt
  - Storage model: JSON file: `data/todos.json` (no database)
  - Initially, load sample data (on first run)
  - API endpoints:
    - GET /api/todos – list all TODOs
    - GET /api/todos/:id – get one TODO
    - POST /api/todos – create
    - PATCH /api/todos/:id – update
    - DELETE /api/todos/:id – delete
  - At the home page show simple API docs: describe endpoints
```

TODO List API in Next.js

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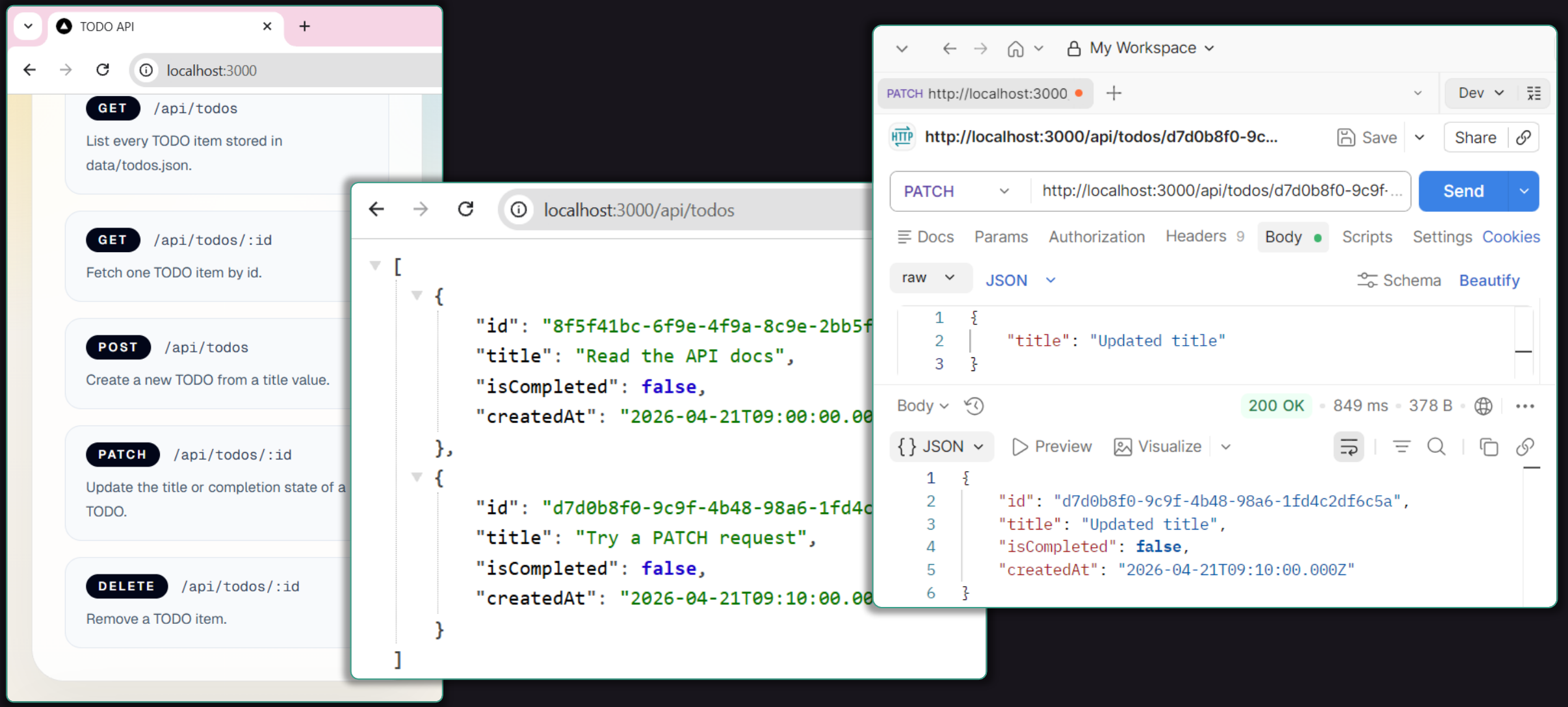

Project Structure

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📁 Folder Structure

app/ ← Next.js pages and API routes
  page.tsx ← home page (API docs)
  api/todos/ ← API route handlers
    route.ts ← GET, POST /api/todos
    [id]/route.ts ← GET, PATCH, DELETE

📦 Supporting Files

lib/
  todos.ts ← data access layer
data/
  todos.json ← JSON storage
package.json ← NPM scripts

🔑 Key Design Principles

- All data access goes through **lib/todos.ts** — no direct file I/O in route handlers
- **Route handlers** stay thin: validate input, call lib, return Response.json(...)
- Keep **todos.json** in the **data/** folder — easy to replace with a real DB later

Testing with Postman

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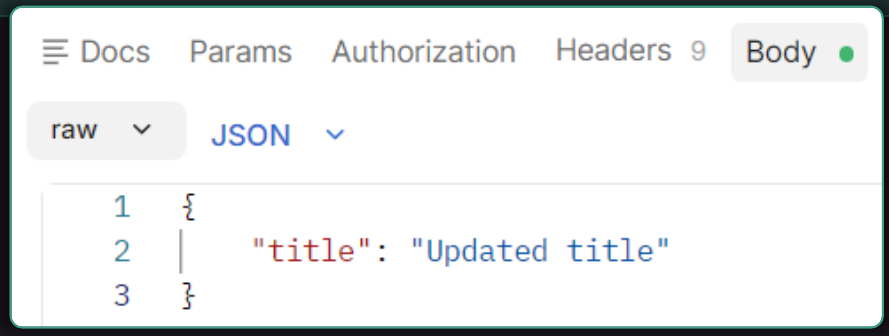
🧪 Postman Tests for "TODO Next.js API"

- Create Postman collection: "TODO Next.js API"
- **GET** /api/todos → 200 OK, JSON array
- **POST** /api/todos + JSON body → 201 Created, new TODO object
- **GET** /api/todos/1 → 200 OK, single TODO
- **PATCH** /api/todos/1 + { "completed": true } → 200 OK, updated TODO
- **DELETE** /api/todos/2 → 200 OK or 204 No Content
- **GET** /api/todos/999 → 404 Not Found, error JSON

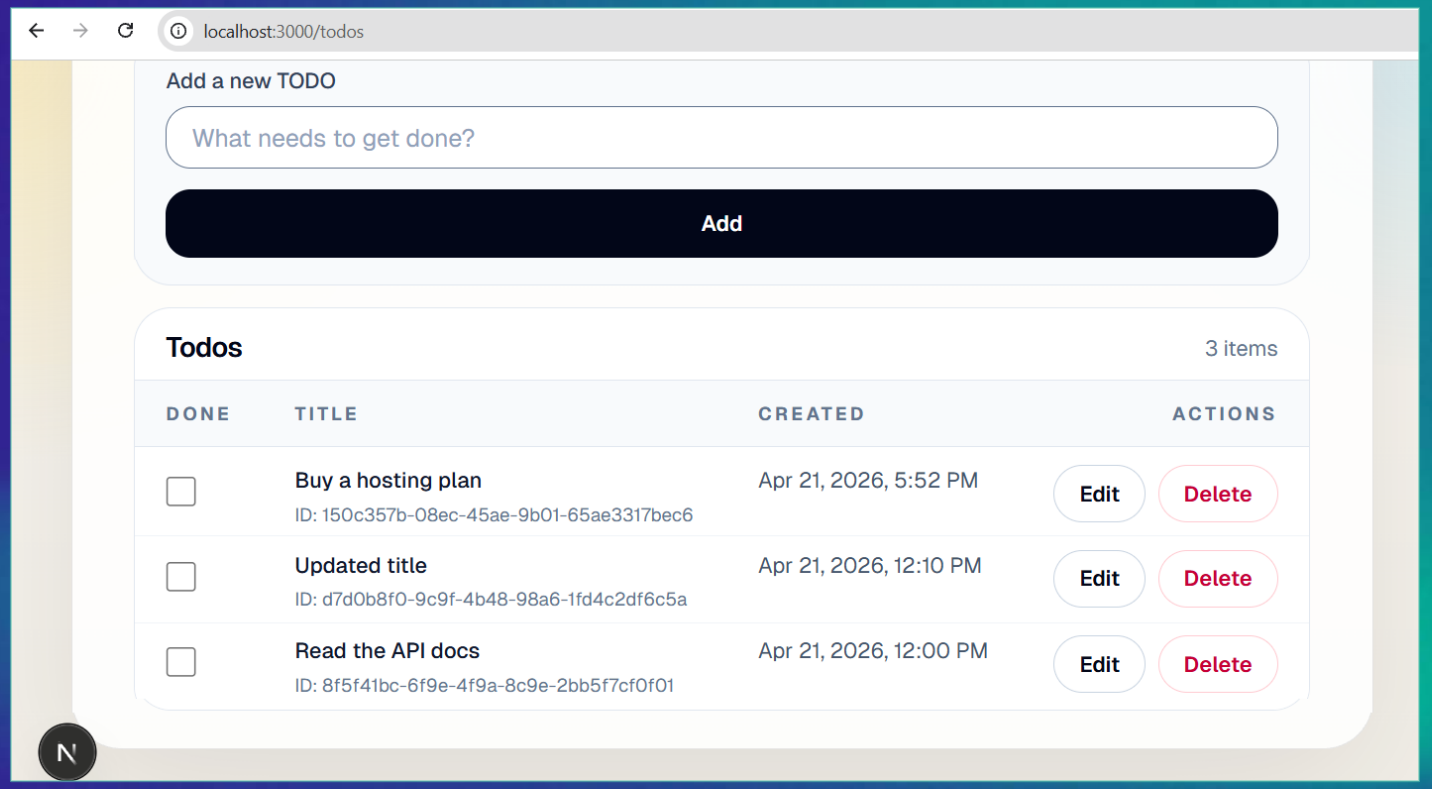
💡 Tips for Postman Testing

- Body → raw → JSON



TODO List Client App

Consume the API from a React UI: Implementing View, Add, Edit, Delete



Next.js Client Apps

- **Next.js** is powerful platform to build **full-stack apps**
- **Server-side** logic (back-end)
  - Runs at the Next.js server, e.g. in Netlify
  - API route handlers, server-rendered React components
- **Client-side** logic (front-end)
  - Traditional client-side React components
  - Client components interact with the server-side logic
- **Mixing** server-side and client-side components

TODO List Client App — AI Prompt

- **AI prompt** to generate a **TODO List client page** in **Next.js**:

```
Implement a TODO List client page in Next.js:
  - A new page: /todos — in the same Next.js project
  - Use the existing API: fetched from `/api/todos`
  - View the list of TODOs — in a table
  - Toggle `Completed` — checkbox for each row
  - Edit title inline — [Edit] button for each row
  - Delete a TODO — [Delete] button for each row
  - Add a new TODO — input field + [Add] button
```

TODO List Client App – Result

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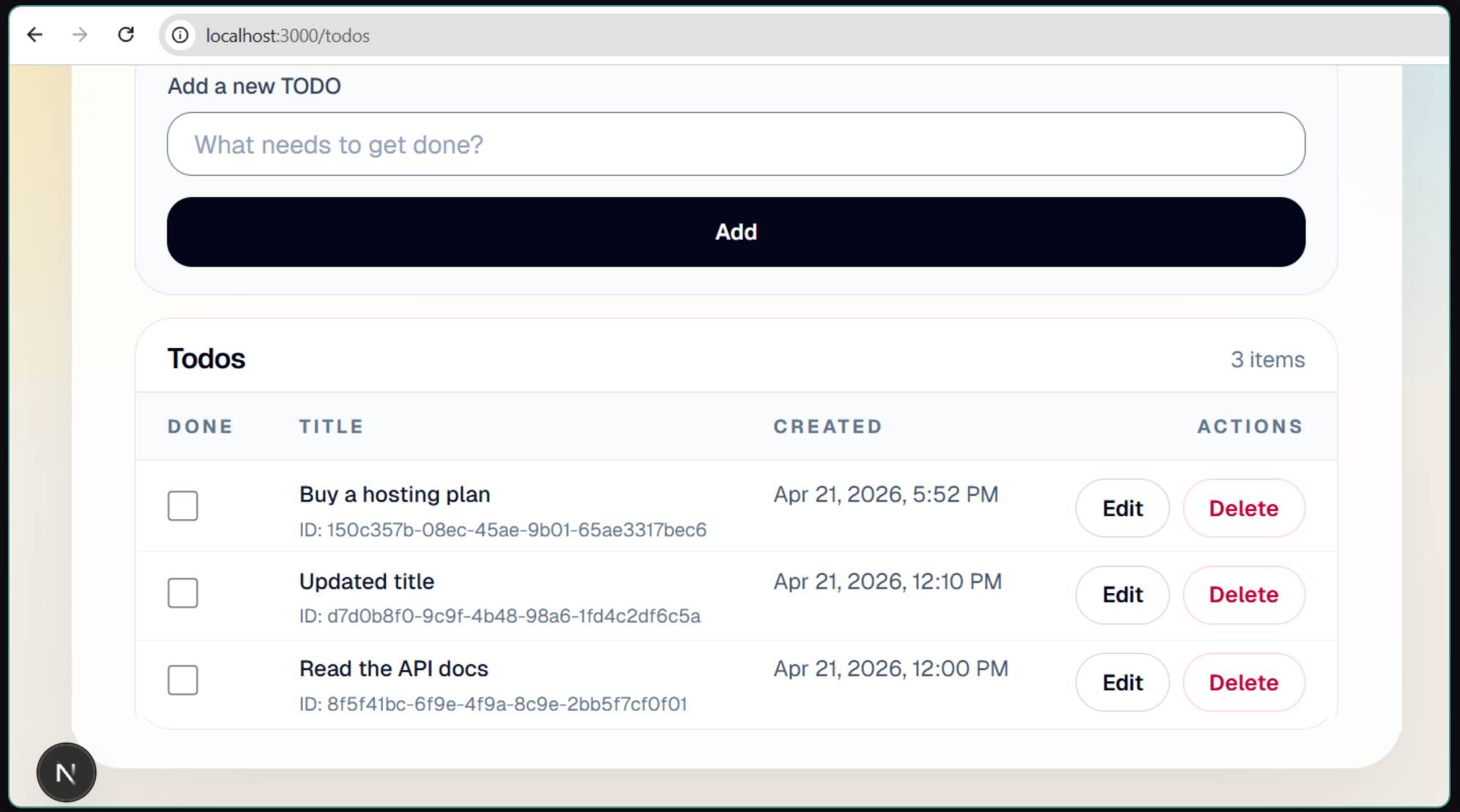

The Client App – Implementation

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🔄 Architecture: Same Project, Two Layers (API + Client)

- Browser → **/todos** page (React client component) → **fetch()** → **/api/todos** (Next.js API route) → **data/todos.json**
- Full-stack in a single project: all code lives in one repo, one `**npm run dev**`
- Later: **add** auth, replace JSON with a **database**, deploy to **Netlify**

⚙️ Technical Implementation

- Client **React component** — must add '**use client**' at the top
- Uses **useState()** to keep the TODOs list locally
- Uses **useEffect()** to fetch todos on page load
- Uses **fetch()** to **POST**, **PATCH**, **DELETE** via the API

Authentication & Authorization

JWT Tokens, Password Hashing

Authentication vs. Authorization

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🔑 Authentication: Who Are You?

- Verify the **identity** of the person or system
- Authentication mechanisms: **email** + **password**, **OAuth**, **magic link**
- Result: a **session cookie** or a signed **auth token** (JWT)
- In our app: **POST /api/auth/login** → returns **JWT** token
- Rule: **auth must happen first** — you can't authorize without identity

🛡️ Authorization: What Can You Do?

- Check **permissions** for the authenticated user
  - Based on **roles**, **scopes**, or resource **ownership**
  - Examples: Is this user an **admin**? Can they delete others' TODOs?
- In our app: only the **owner** can see/edit their own TODOs
- Rule: **authenticate** first, then **authorize** for each action

JWT Tokens

- **JWT** (JSON Web Token) is modern standard for **authentication** and **authorization**
  - A JWT token holds info about the **currently logged-in user**: **user_id** / **username**, **roles** and **permissions** (optional)
  - Holds a **digital signature**, created by the token issuer
- The **JWT token** is:
  - **Issued at login** (after credentials are verified)
  - **Sent with each request** to identify and authorize the user
  - **Removed** from the client at **logout** (to forget logged-in user)

What is a JWT Token?

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🔐 JWT Structure

- JWT = **JSON Web Token** (RFC 7519)
- A self-contained, cryptographical token
- Three parts separated by dots (**.**):
  - **Header.Payload.Signature**
- Example: **eyJhb... . eyJzW... . n2Hw...**

📦 JWT Payload (Decoded)

- The payload contains 'claims' about the user:

  { **"sub": "123", "email": "peter@email.com",**
  "iat": 1713350400, ← issued at
  "exp": 1713436800 ← expires at
  }

- Anyone can **decode JWT** — never store secrets in it!

✅ Why JWT?

- **Stateless** — the server doesn't need to store session state (works great for serverless)
- **Self-contained** — the token carries all necessary information about the user
- **Portable**, **standard** — works across Web apps, mobile apps, and third-party clients

Inside JWT Tokens

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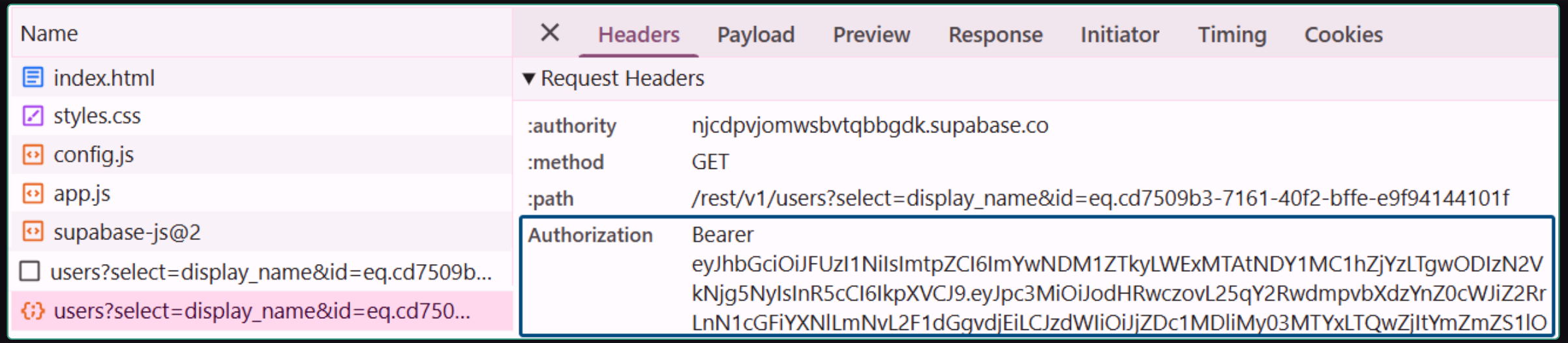
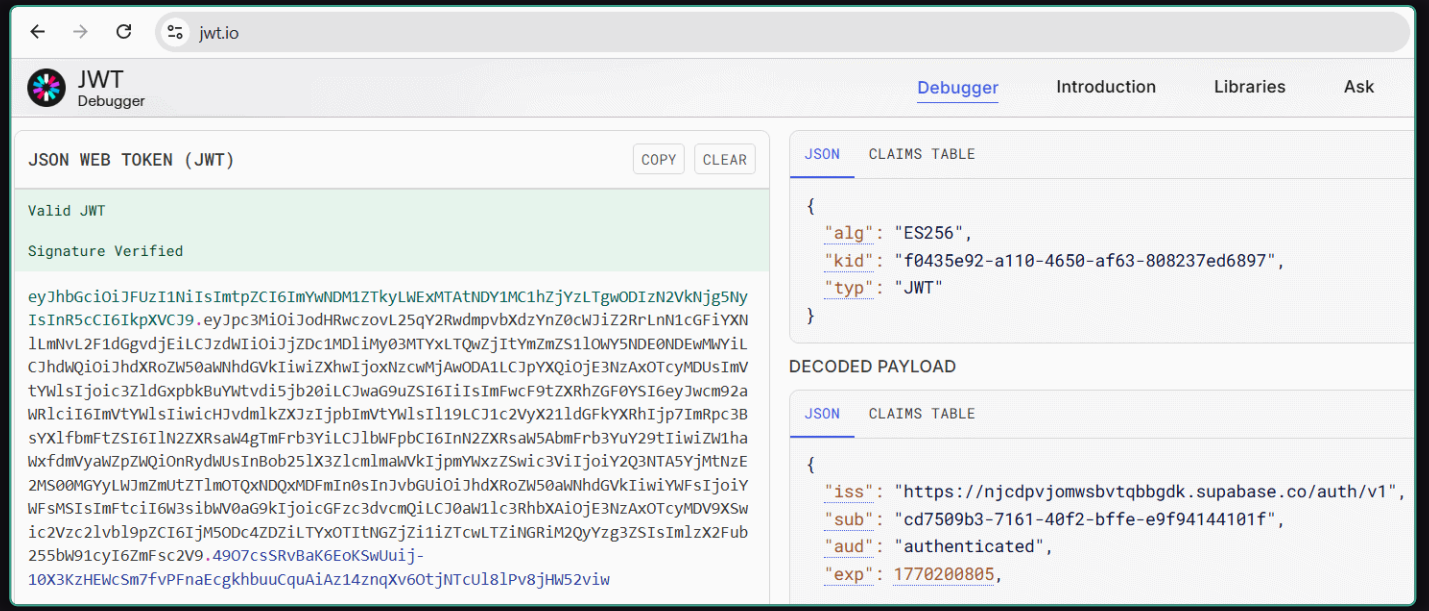
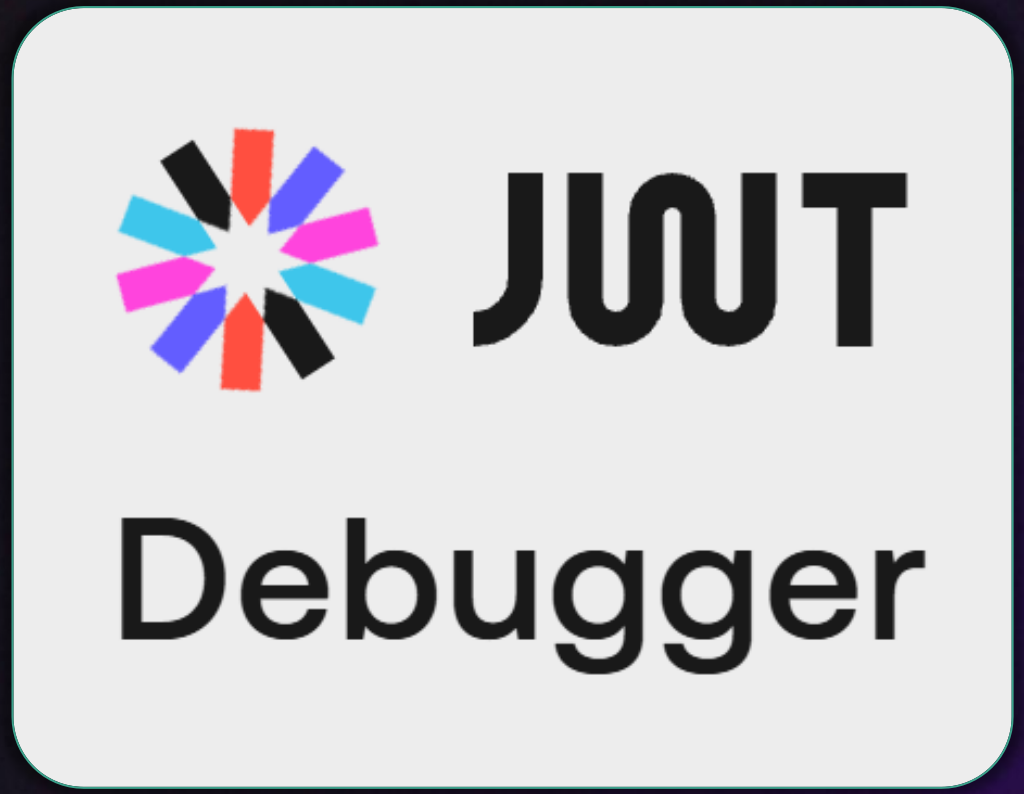


header	payload	signature
{ "alg": "ES256", "typ": "JWT" }	{ "sub": "jonny86", "iat": 1516239022 }	KMUFsIDTnFmyG3nMiGM6H9FNFUROf3wh7S...





JWT Tokens

Live Demo

https://www.jwt.io

RESTful APIs with Auth

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🔑 Typical Authentication Flow (Auth)

1. Client sends **auth claims** (username + password): **POST /api/auth/login**
2. Claims are **checked** by the server: OK → issue **JWT** token. Not → **error**
3. The **JWT token** is saved by the app for the next requests

🔄 Typical Request Flow with Auth (Authorized Requests)

1. Client sends request with **Authorization: Bearer <token>** header
2. Server extracts the **token** and **verifies** the signature
3. Server reads the **user ID** from the token payload
4. Server **checks permissions**: is this user allowed to do this action?
5. If **OK** → process the request. If not → return **401** or **403**

Hashing Passwords

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⚠️ Never Store Plain Text Passwords!

- Storing **passwords as plain text** is a **critical security flaw**
- Always hash passwords with a **one-way algorithm**:
  - **bcrypt** — most common (npm: **bcryptjs**)
  - **argon2** — modern, for new projects
- Store only the **hash** + **salt** — the password is never stored

💡 Salt Rounds Parameter

- **bcrypt.hash(password, 10)**
- 10 == cost factor (salt rounds)
- Higher rounds = stronger hash = slower computation
- 10 is a good default

🔧 BCrypt: Register User

```
const hash = await
  bcrypt.hash(password, 10);
// Save hash to users.json / database
```

🔧 BCrypt: Login User

```
const ok = await
  bcrypt.compare(password, hash);
if (!ok) return 401 (Unauthorized)
```

BCrypt Hashes

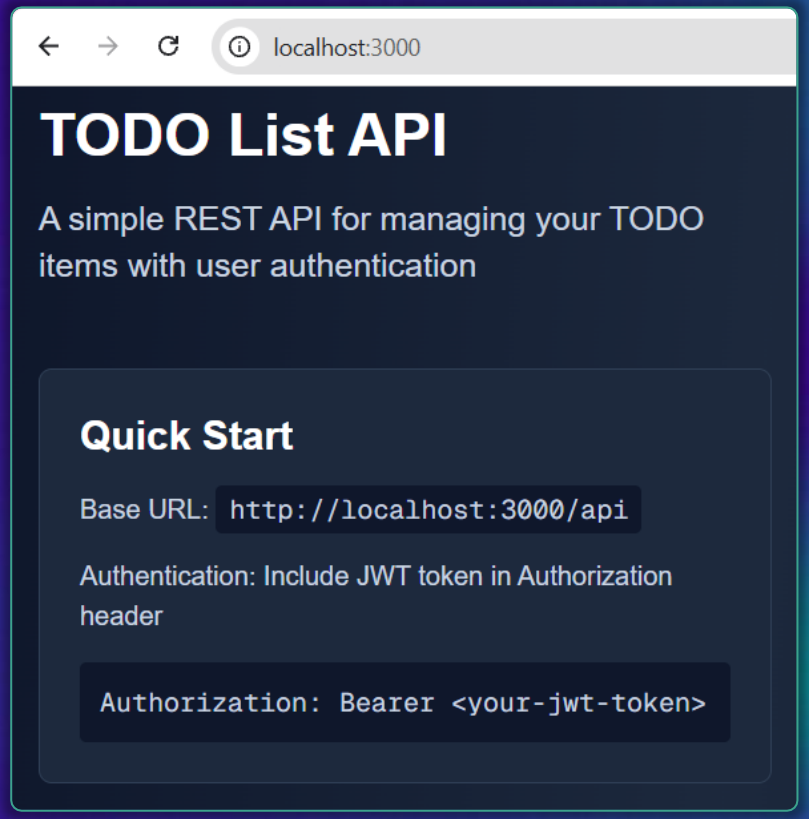
https://bcrypt-generator.com

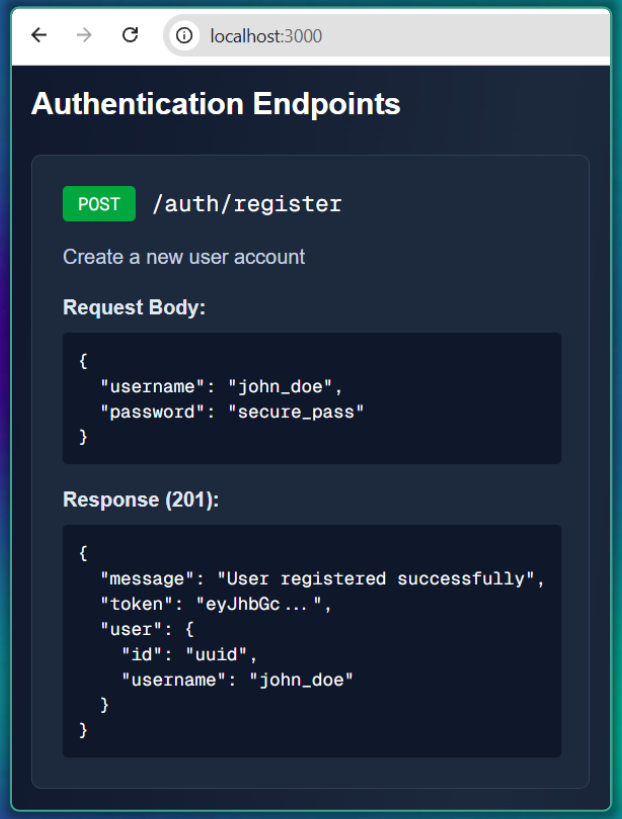
```
$2a$10$EG5S0thG2oaPBzXyfK/qS.J
Aqc1gawd4Ik5O8SdcjIjUgZLSkd7Wi
```

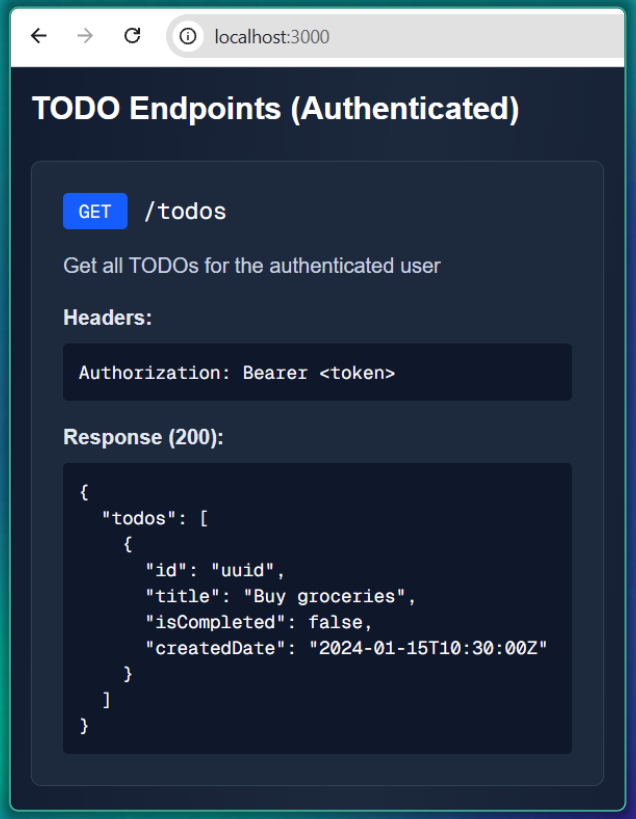
Live Demo

TODO List API with JWT Auth

Register, Login, Logout, Protected Endpoints







TODO List API with JWT Auth

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- We want to build a **TODO List API server** with **auth**
  - Users **register** / **login** and keep their **own TODO items**
  - Auth endpoints: **register** / **login** → return **JWT token**
  - Data endpoints: **list** / **view** / **create** / **edit** / **delete** TODO
    - Authenticate with a JWT token
  - Data model:
    - **Users**: username, password hash, TodoItem[]
    - **TODO items**: title, isCompleted, createdDate

TODO List API with JWT Auth in Next.js

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- AI prompt to build **TODO List API** with **JWT Auth** in **Next.js**

Implement a **TODO List API server** as simple **Next.js** app:
- Users **register** / **login** and keep their **own TODO items** on the server
- Auth endpoints: **register** / **login** → return **JWT token**
- Data endpoints: **list** / **view** / **create** / **edit** / **delete** TODO (for authenticated users, with JWT token)
- Data model: keep data in a simple file `**data/app-data.json**`
  - **Users**: username, password hash (bcrypt), TodoItem[]
  - **TODO items**: title, isCompleted, createdDate
- Generate simple **API docs** at the home page: describe the endpoints

Security Warning: JWT_SECRET

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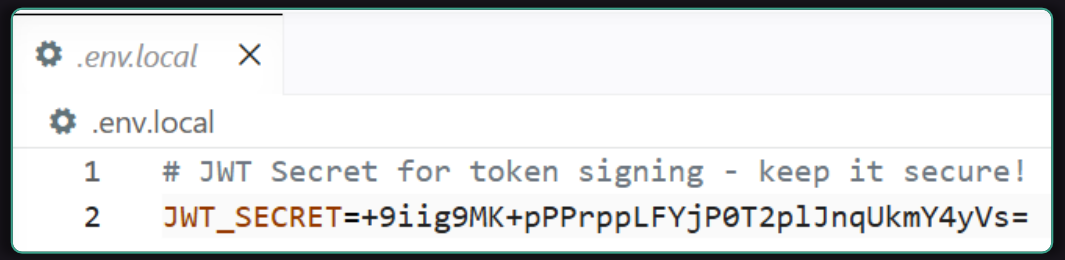
- Most apps use an environment config value **JWT_SECRET**:

`.env.local` *(your private config file)*

```
JWT_SECRET=random_long_secret
```



- Critical **security flaw**: if **JWT_SECRET** leaks, attackers can **login without a password**!
- AI prompt to **configure the JWT_SECRET**:

Generate a **random `JWT_SECRET`** in **`.env.local`**

- Even better: configure your key **manually** (without AI agent)

TODO List API with JWT Auth

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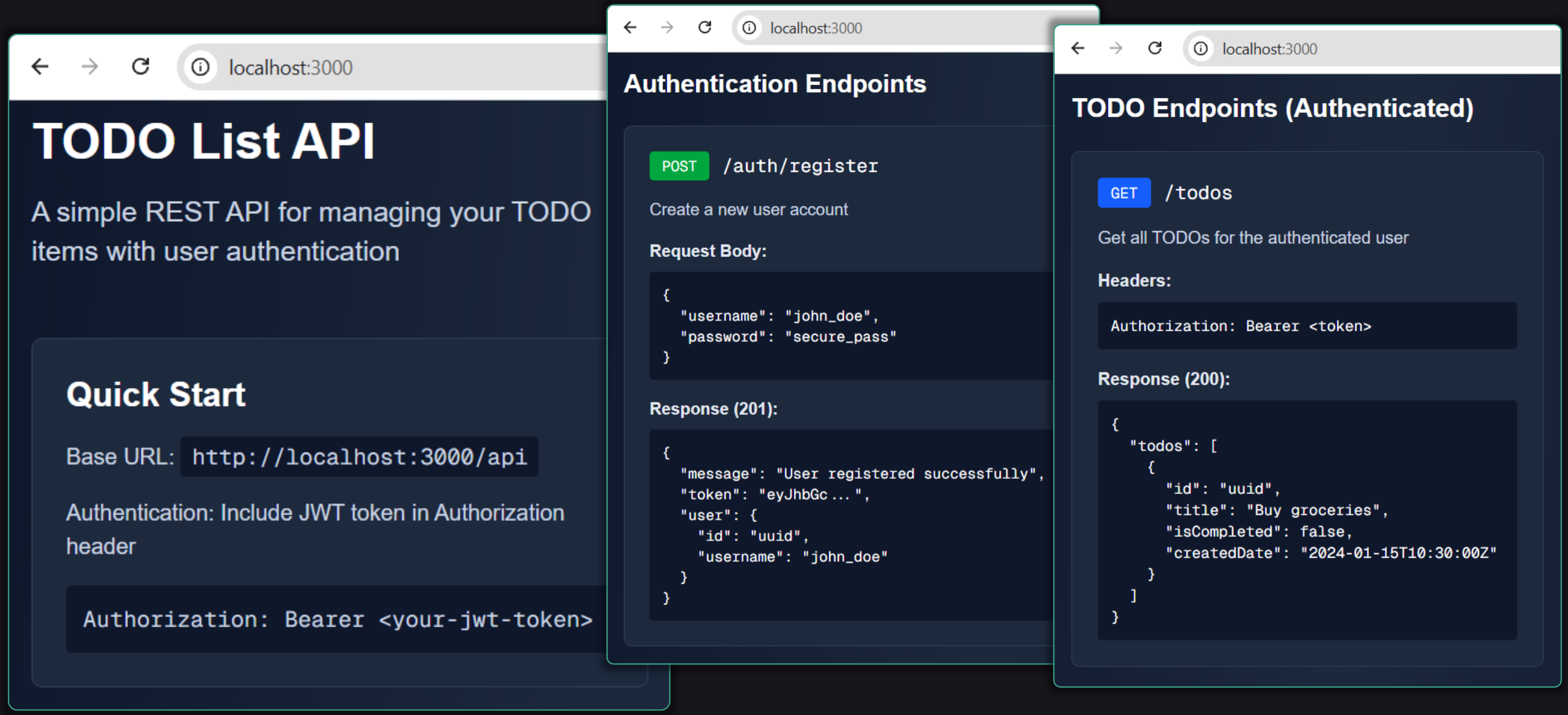

Inside the TODO List API with Auth

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🆕 Auth Endpoints

- **POST /api/auth/register** — create new user, return JWT
- **POST /api/auth/login** — verify credentials, return JWT
- **Logout** — no endpoint at the server (client drops token)
- **GET /api/auth/me** — return current user from JWT
- Storage: **data/app-data.json** (email + passwordHash)

🔒 Protected Data Endpoints

- **GET**, **POST /api/todos**
- **GET**, **PATCH**, **DELETE /api/todos/:id**
- All **/api/todos** endpoints now require **JWT auth**
  - **Authorization: Bearer <jwt_token>**
  - Missing or invalid token → **401 Unauthorized**
- Each user sees only their own TODOs
  - JWT token holds the **user ID**

Testing Auth with Postman

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📋 Step 1 — Register

- **POST http://localhost:3000/api/auth/register**
  - Body: **{ "username": "steve", "password": "pass123" }**
  - Expected: **201 Created** + **{ "token": "eyJ..." }**
- Copy the **token** for the next requests
- Try registering the same username again → **409 Conflict**

📋 Step 2 — Use the Token

- **GET http://localhost:3000/api/todos**
  - **Authorization: Bearer eyJ...**
  - Expected: **200 OK** + **{ "todos": [] }**
- Without the Authorization header:
  - Expected: **401 Unauthorized** + { error: 'Unauthorized' }

📋 Step 3 — Full Auth Flow

- **POST /api/todos** with token → **201 Created** (TODO created for this user)
- **GET /api/todos** with same token → **200 OK**, shows your TODOs
- Login as a DIFFERENT user → **GET /api/todos** → sees empty list (own TODOs only)

Testing Auth with Postman: Demo

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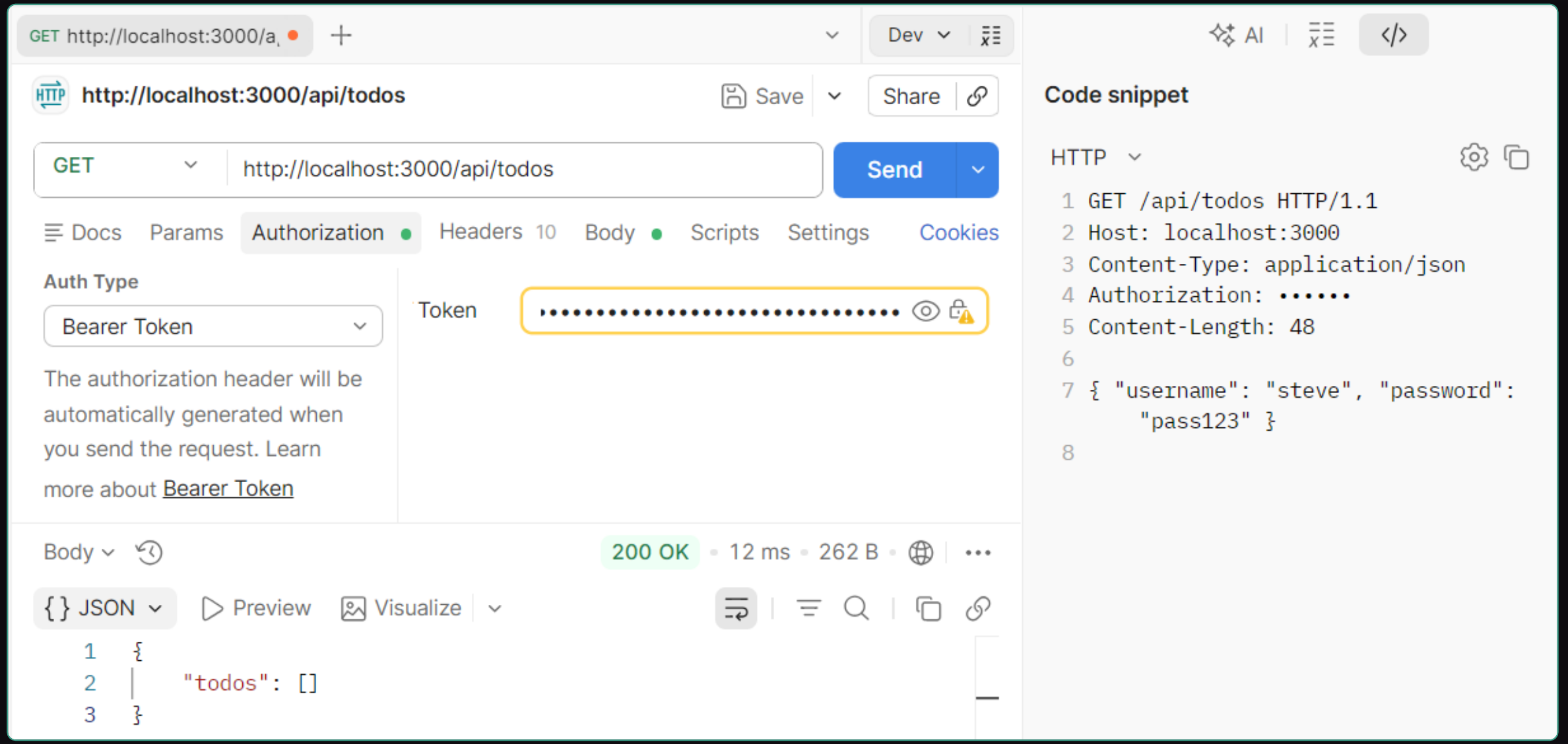
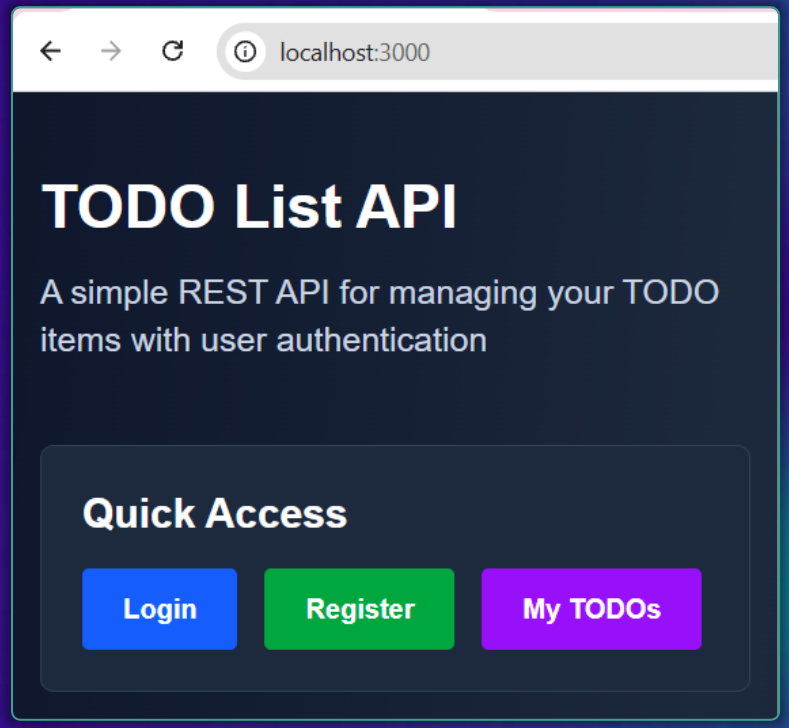
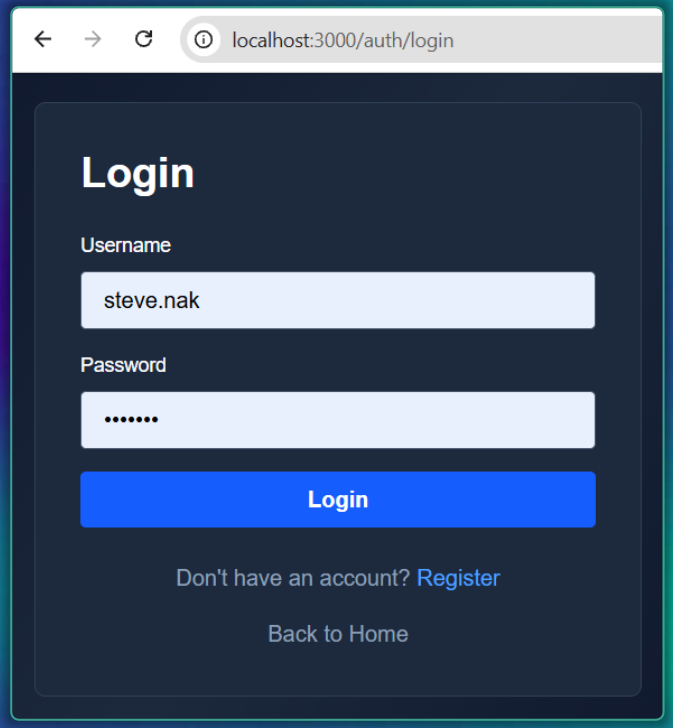
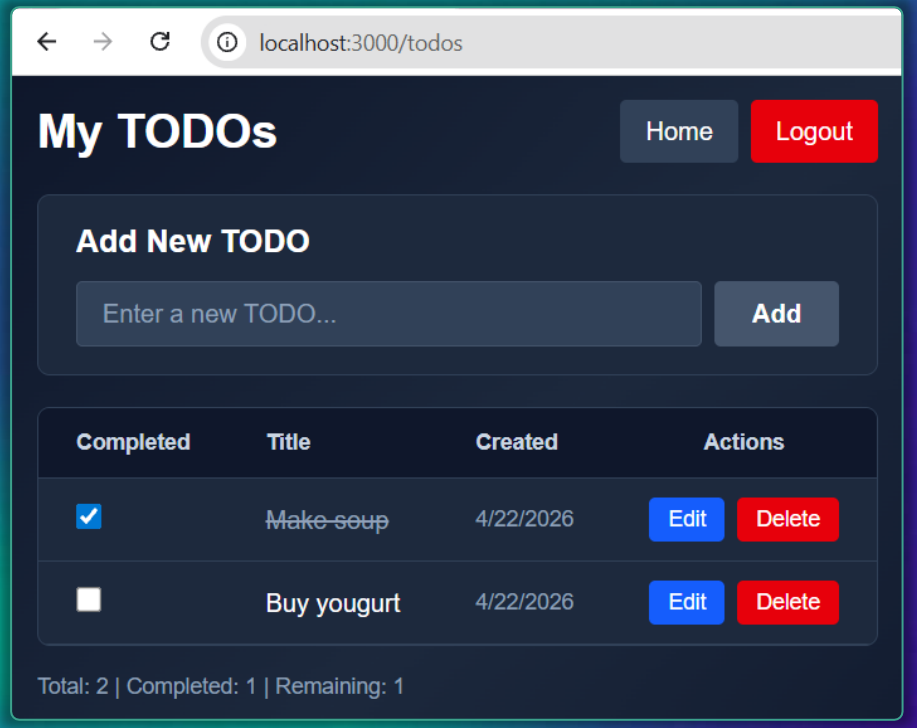


TODO List Client App with Auth

Register / Login UI, Storing Auth Tokens, Protected Routes







TODO List Client App — AI Prompt

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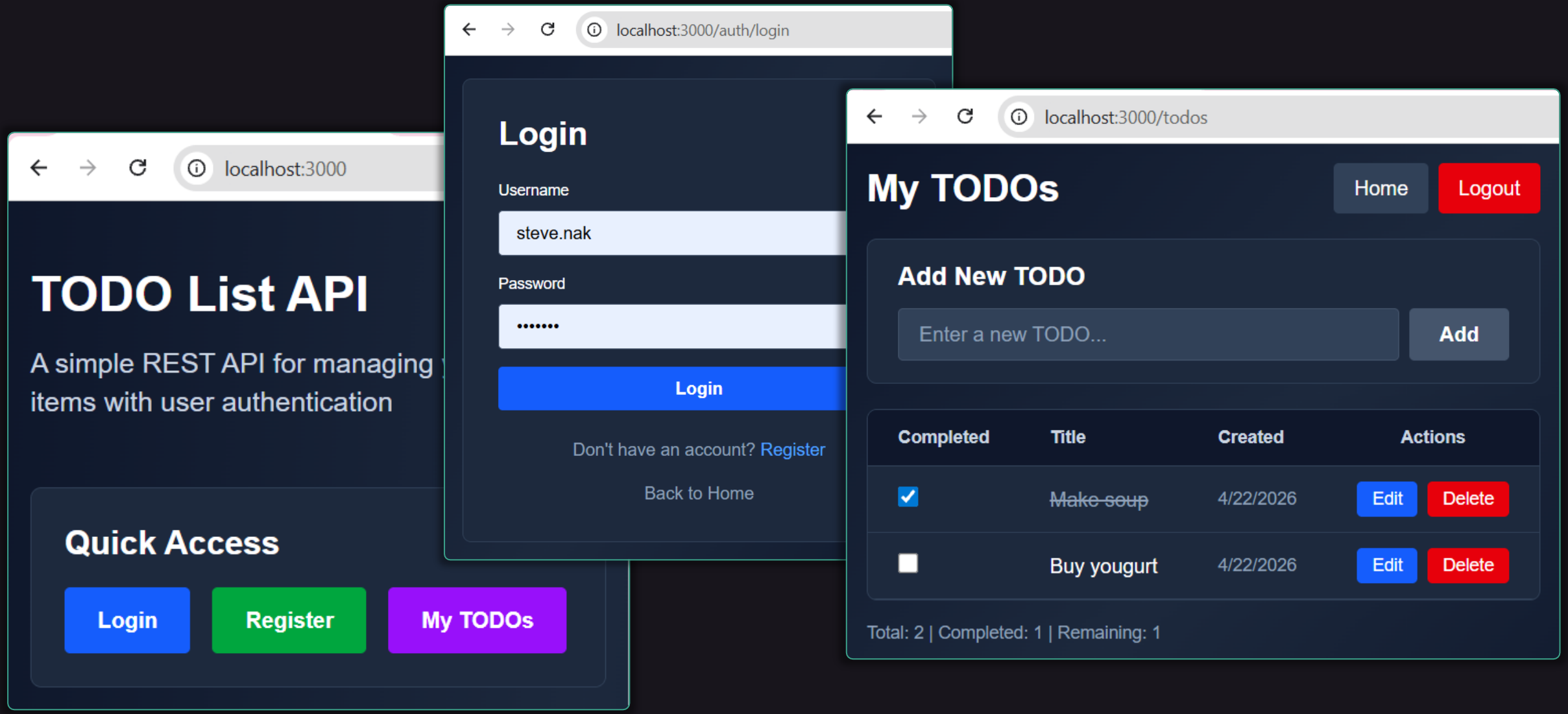
- **AI prompt** for **TODO List client app** with **auth** in **Next.js**:

```
Implement a TODO List client app in Next.js:
  - Implement auth: login / register / logout
  - A new page: /todos – in the same Next.js project
    - View the list of TODOs – in a table
    - Toggle `Completed` – checkbox for each row
    - Edit title inline – [Edit] button for each row
    - Delete a TODO – [Delete] button for each row
    - Add a new TODO – input field + [Add] button
  - Use the existing API: fetched from `/api`
  - App Home page: links to auth / TODOs / docs
```

TODO List Client App – Result



The Plan – Client App with Auth

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📄 Auth Pages

- **/auth/register** — register form
  - Username + password + [Register] button
  - **POST /api/auth/register**
- **/auth/login** — sign in form
  - Username + password + [Login] button
  - **POST /api/auth/login**

🔒 Protected /todos Page

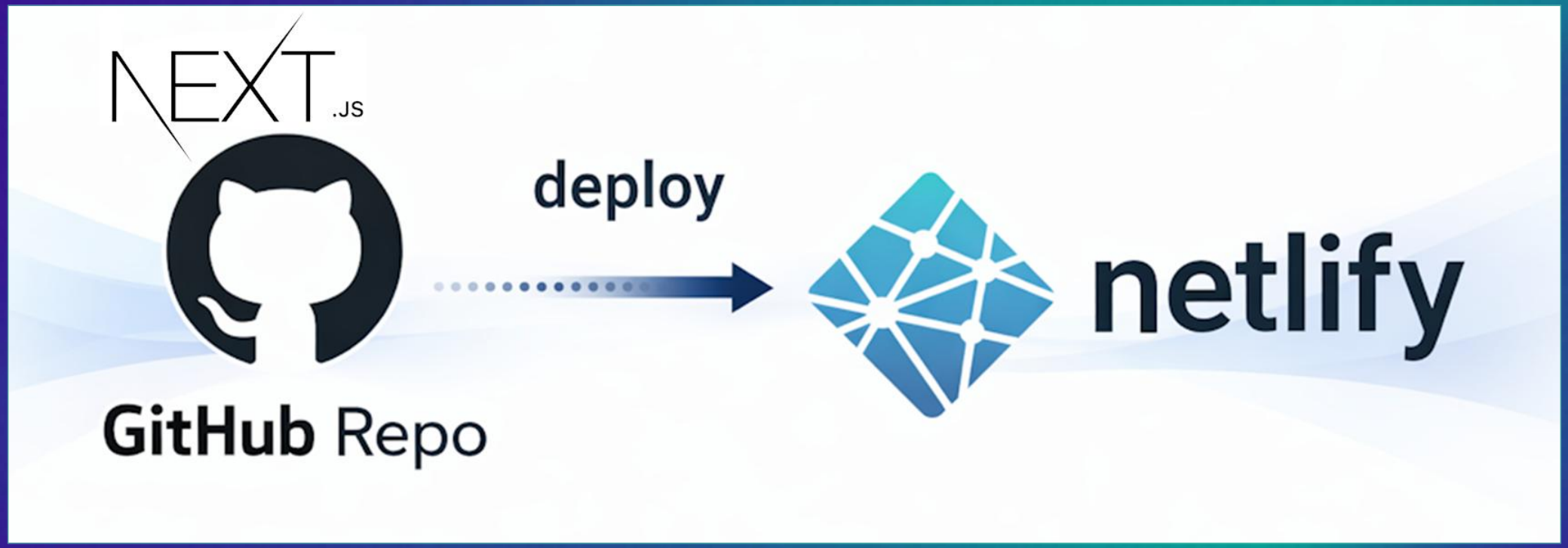
- **Login**: store the **JWT token** in **localStorage**
- On **/todos** mount: check for **token**
  - **Load TODOs** if token is available
  - Redirect to **/auth/login** if missing
- Send Authorization: **Bearer <token>** on API calls
- **Logout**: clear **localStorage** token + go to **/login**
- Top nav: links to **/todos**, **/login**, **/register**, **[Logout]** button

🔒 Security Warning

- **Auth token** in **localStorage** might be attacked with **XSS** (Cross-Site Scripting)

Deploy Next.js API to Netlify

Publish to GitHub, Deploy Next.js App to Netlify, Serverless Limitations



Deploy an Next.js API to Netlify

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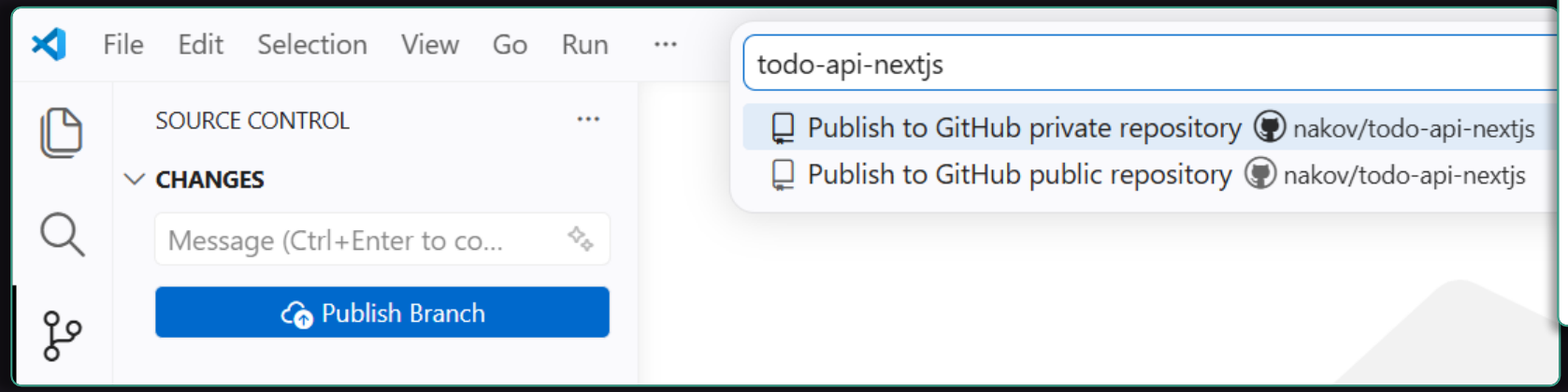
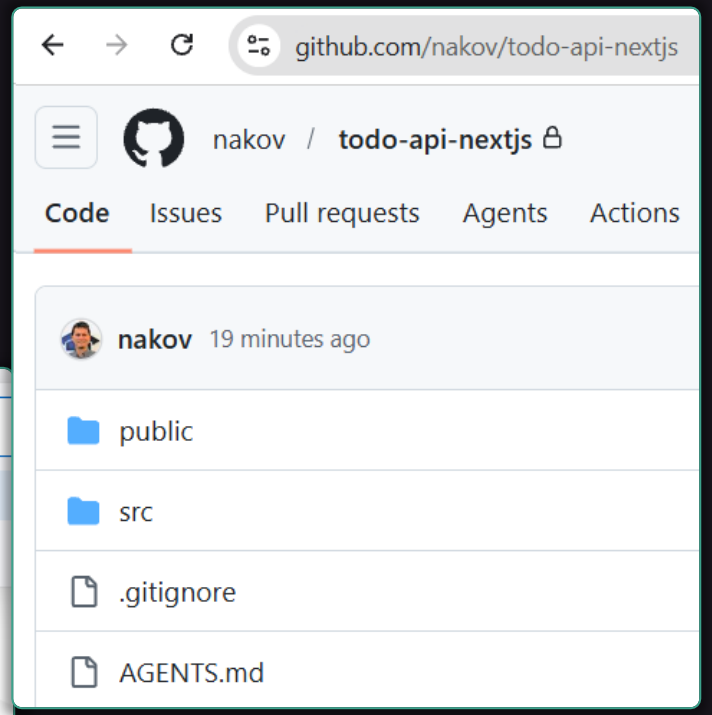
- How to **deploy a Next.js API** project to **Netlify**?
  1. **Publish** your project to **GitHub**
  2. **Deploy to Netlify** from GitHub
- Publish to **GitHub** from **VS Code**





  - By default, `**.gitignore**` will skip `**.env.local**` (config secrets), also folders **node_modules/** and **.next/**

Deploy to Netlify from GitHub

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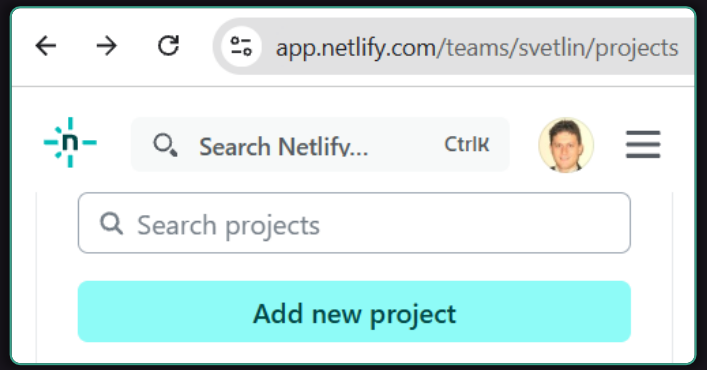
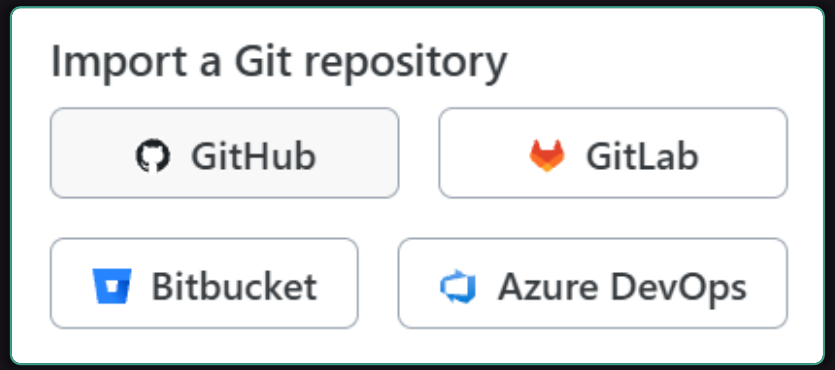
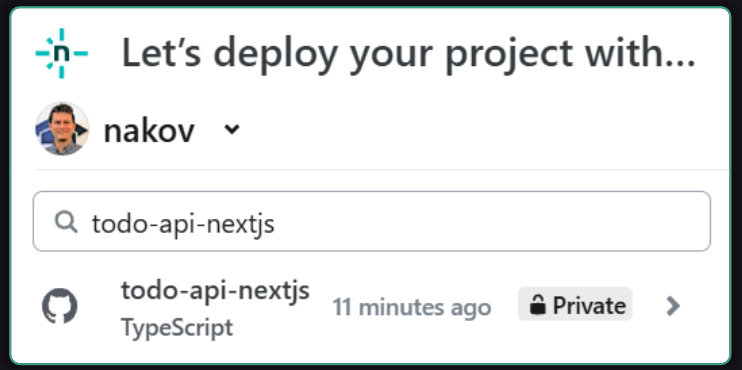
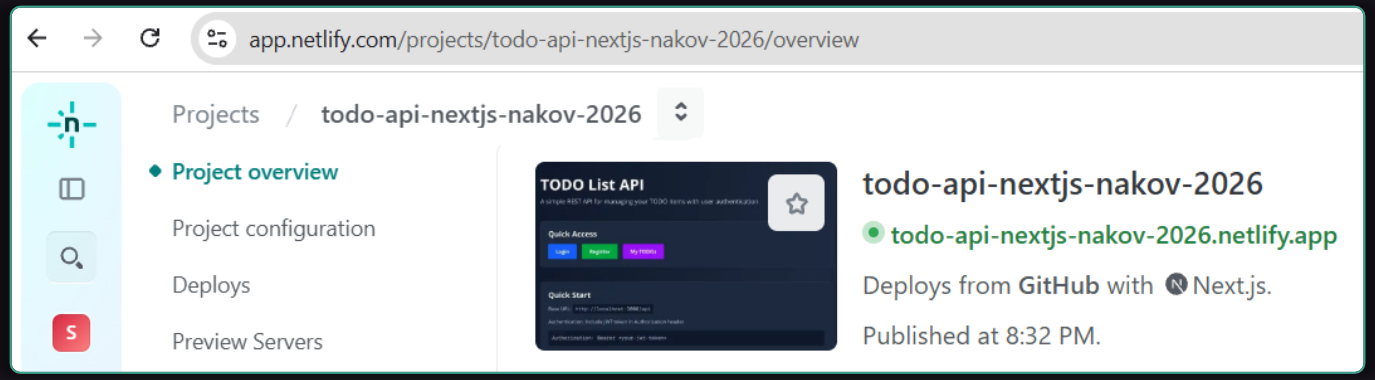
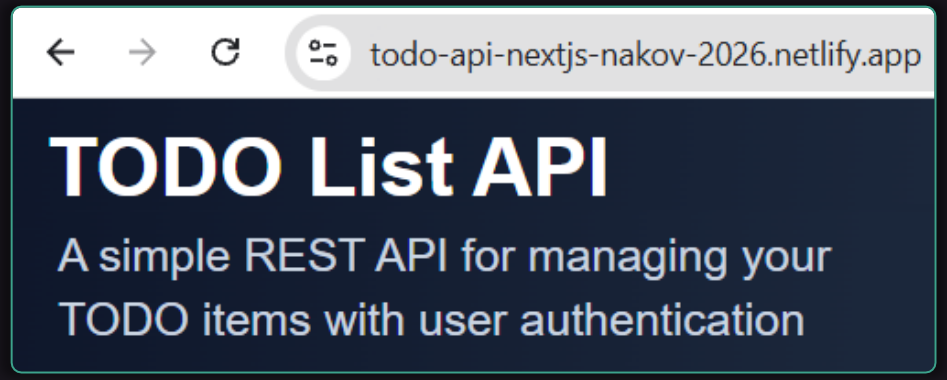
- **Deploy to Netlify** from GitHub
  - Netlify Dashboard → [Add New Project] → [GitHub] → Authorize → Select Project → Configure → Deploy











The App Works in Read-Only Mode!

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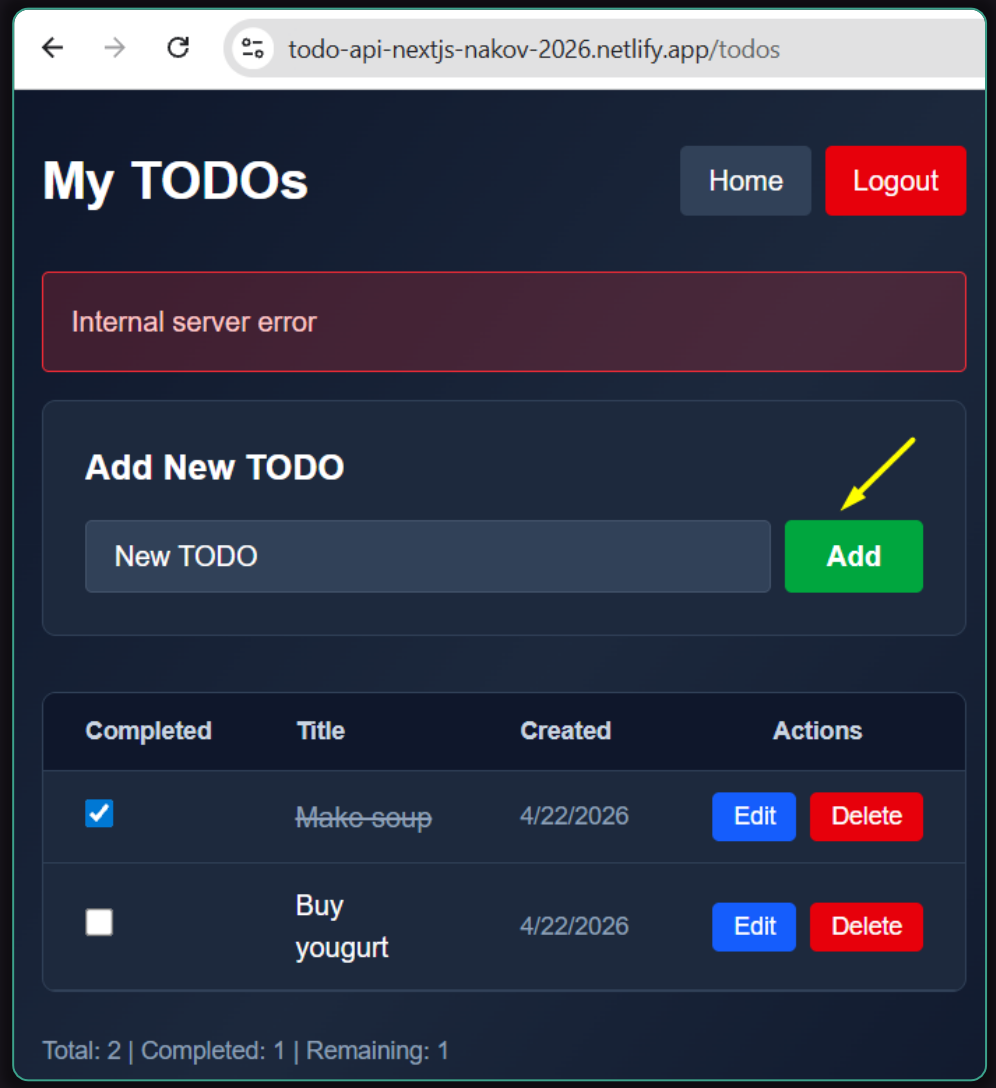

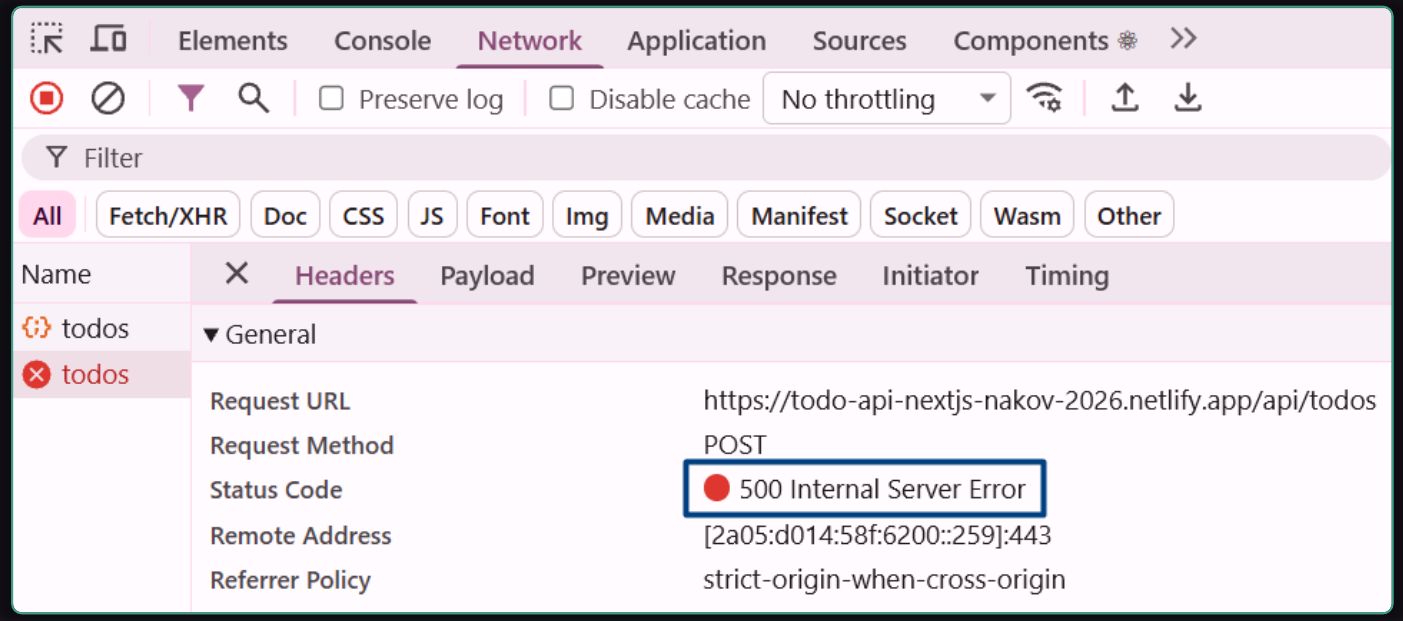

- Netlify has **read-only** file system
  - API **read** requests will **work**, but API **write** requests will **fail**

Netlify – A Serverless Environment

- **Netlify** is a **serverless** deployment platform
  - In Netlify app's `**data**` folder in file system is **read-only** (by design)
- How **serverless** platforms work?
  - Requests run on-demand, **without preserving state**
  - **Request** → **server** (from the pool) → run the **code** → return **result**
  - Each request may be served by different runtime (machine)
- How to preserve state?
  - Use a **database** (PostgreSQL), **cache** (Redis), or **external** API
  - Don't persist data in **local files** (even in **/tmp**, which is writable)

How Serverless Actually Runs?

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⚡ Per-Request Lifecycle

1. A **request** arrives at Netlify's edge network
2. A new **Node.js process** is spawned (or reused from a warm pool)
3. **Route handler runs** → returns a **response**
4. The **process** may be **stopped** shortly after (cold → warm → cold)
   - No guarantee of which **host machine** handles the next request

🚫 Serverless Constraints

- **Memory** does **NOT persist** between requests
- **File system** is **read-only** (our case)
- **/tmp** is writable — but **temporary**, not shared across instances
- No long-running **background tasks**
- Concurrent requests may run on **different host machines**

🖥️ Serverless vs. Server-Based Hosting

- **Server-based** (Railway, Render): long-running process, persistent memory, writable filesystem
- **Serverless** (Netlify, Vercel, AWS Lambda): non-persistent, scales to zero, pay-per-request
- For persistent data in serverless: you need an **external database** — not a local file

What We Learned Today

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- **Client-Server Model & HTTP** — browsers send HTTP **requests**; servers return status codes and JSON **responses**
- **RESTful API Design** — based on **resources**, addressed by **URL**, using **HTTP** verbs (GET / POST / PATCH / DELETE), status codes and **JSON**
- **Postman** — a powerful developer tool for sending **HTTP requests** and inspecting the **responses**
- **Next.js API Routes Handlers** — building a **RESTful APIs** with TypeScript inside a Next.js project (route handlers in the **/app/api/** folder)
- **JWT Authentication** — server issues **JWT tokens**, encapsulating signed user data; client sends tokens back on each request for authentication
- **Deploying to Netlify** — Next.js projects run natively on Netlify (deploy from GitHub repo) but: **non-persistent** memory, **read-only** file system

Questions?

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